



Supplementary Materials for

Physical mixing of a catalyst and a hydrophobic polymer promotes CO hydrogenation through dehydration

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Materials

Tetraethylorthosilicate (TEOS, >99%), divinylbenzene (DVB), azobis isobutyronitrile (AIBN), *n*-hexane (>99%, GC), polystyrene (PS) and SiO₂ (AR) were obtained from Aladdin Chemical Reagent Company. Diethoxydimethylsilane (DEMS, 99%), ethanol (AR, >99.5%), methanol (AR, >99.5%), Co(NO₃)₂·6H₂O, 50 wt% Mn(NO₃)₂, CuSO₄·5H₂O (99%), ethyl acetate (EA), dimethyl formamide (DMF), and Na₂CO₃ were obtained from Sinopharm Chemical Reagent Co. Ltd. Graphite (Gra) and sodium styrene sulfonate (SSS) were purchased from Shanghai Macklin Biochemical Co. Ltd. Polytetrafluoroethylene (PTFE) was purchased from Adamas Reagent Ltd. Feed gas (containing pure CO, pure H₂, CO/Ar, H₂/CO/Ar, H₂/Ar, C₂H₄, C₃H₆, and CO₂ etc.) were supplied by Hangzhou Jingong special gas Co. Ltd.

Catalyst preparation

Synthesis of CoMnC. The CoMnC catalyst was prepared by a co-precipitation method described previously (8). Typically, 29.4 g of Co(NO₃)₂·6H₂O and 17.9 g of 50 wt% Mn(NO₃)₂ were dissolved in 75 mL of deionized water to obtain solution A (mole ratio of Co/Mn at 2/1). Meanwhile, 24.0 g of Na₂CO₃ were dissolved in 75 mL of deionized water to obtain solution B. Then solutions A and B were simultaneously dropwise added into a beaker containing 200 mL of deionized water under vigorous stirring at 30 °C. In this process, the pH value was maintained at ~8. The obtained suspension was kept with stirring for 1 h and aged for 2 h at 30 °C. After the precipitate was centrifuged, washed with a large amount of deionized water, dried at 100 °C for 10 h, and calcined at 330 °C for 3 h, the CoMn oxide was obtained. By the pre-reducing treatment and catalyzing syngas conversion, the CoMn oxide could be transferred into CoMn carbide (CoMnC).

Synthesis of nonporous PDVB. 0.5 g of AIBN was added into 10 g of divinylbenzene with stirring at room temperature for 1 h, then the liquor was transferred into an autoclave and thermally treated at 100 °C for 24 h. The obtained solid was washed with methanol and dried at 100 °C overnight to obtain the nonporous polydivinylbenzene, which was abbreviated as PDVB.

Synthesis of nanoporous PDVB. 10 g of divinylbenzene and 0.5 g of AIBN was dissolved in 50 g of solvent under stirring at room temperature for 1 h, then the liquor was transferred into an autoclave and thermally treated at 100 °C for 12 h. The obtained solid was washed with methanol and dried at 100 °C overnight to obtain the nanoporous polydivinylbenzene. In this process, the DMF and EA were used as solvents, and the obtained samples were denoted as PDVB-DMF and PDVB-EA, respectively. In this manuscript, if there was no special notification, the PDVB means the nonporous sample, while the PDVB-DMF and PDVB-EA were denoted as the nanoporous samples.

Synthesis of PDVB-SSS. As a typical run for synthesis of PDVB-SSS with a molar ratio of SSS/DVB at 0.03, 0.5 g of AIBN was added into 9.5 g of DVB and 0.5 g of SSS with stirring at room temperature for 1 h, then the liquor was transferred into an autoclave and thermally treated at 100 °C for 24 h. The PDVB-SSS was obtained by washing with methanol/ethanol and drying at 100 °C overnight. For comparison, the PDVB-SSS materials with molar ratios of SSS to DVB at 0.07, 0.16, 0.27, 0.42 were synthesized from similar procedures except for using 9.0 g of DVB and 1.0 g SSS, 8.0 g of DVB and 2.0 g SSS, 7.0 g of DVB and 3.0 g SSS, 6.0 g of DVB and 4.0 g SSS in the starting mixture, respectively. Note: the SSS could not be completely dissolved in the starting DVB liquor, but this method can form a relatively homogeneous mixture of hydrophobic and hydrophilic individuals in the final materials for varying the wettability.

Synthesis of SiO₂-Me. 7.49 g of TEOS and 0.592 g of DEMS were dropped into a mixed liquor containing 80 mL of ethanol, 100 mL of water, and 6 mL of aqueous ammonia. After stirring at room temperature for 12 h, the liquor was removed by distillation under vacuum to get the solid powder, which was then dried at 100 °C overnight to obtain the amorphous silica modified with methyl groups (SiO₂-Me).

Synthesis of CoMn@Si and CoMn@Si-c. The CoMn@Si was prepared by the modified Stöber method (4). Typically, 3 g of the prepared CoMn oxide was dispersed into 300 mL of ethanol by ultrasonication. Then, 15 mL of TEOS was dropped into the mixture with stirring at room temperature for 4 h, followed by adding 30 mL of ammonia and 120 mL of water. After stirring the mixture at room temperature for another 4 h, the solid product was filtrated, washed with a large amount of ethanol, and dried at 100 °C overnight. Finally, the CoMn@Si sample was obtained. The hydrophobic encapsulation of CoMn@Si was achieved by further modification using organosilane. Typically, 1.5 g of CoMn@Si was pre-dried at 150 °C for 12 h and dispersed into 10 mL of *n*-hexane under stirring at room temperature. Then, 3.0 g of DEMS was continuously dropped into the mixture for 1 h under stirring, followed by an ultrasonic treatment for 3 h. The sample donated as CoMn@Si-c was obtained by filtrating, washing with *n*-hexane, and drying at 100 °C overnight.

Catalyst characterization

Powder X-ray diffraction (XRD) patterns were collected on a Rigaku D/MAX 2550 diffractometer with CuK α radiation ($\lambda = 1.5418 \text{ \AA}$). Nitrogen sorption isotherms were measured using a Micromeritics ASAP 2020M system. Transmission electron microscopy (TEM) images were recorded on a JEM-2100F electron microscope (JEOL, Japan) with an acceleration voltage of 200 kV. Scanning electron microscopy (SEM) experiments were performed with Hitachi SU-8010 electron microscopes. Metal loadings and Si/Al ratios of the catalysts were determined by inductively coupled plasma analysis (ICP, Perkin-Elmer 3300DV). IR characterization was

performed with Thermo Nicolet NEXUS 470 FT-IR system. Thermogravimetric curves (TG) were performed on a SDT Q600 Simultaneous DSC-TGA in flowing air with a heating rate of 10 °C/min. TG-MS profiles were obtained from a METTLER TGA/DSC instrument connected to a mass spectrometer (LCD200M, TILON). ¹³C NMR spectra were obtained from a Bruker Avance III HD 400 MHz spectrometer. Water-droplet contact angles were performed with Belsorp MaX II system. Water adsorption tests were performed with video-based contact angle measuring device (OCA 20, Dataphysics Co. Ltd).

CO-hydrogenation *in-situ* FTIR spectra were recorded using a Nicolet is50 FT-IR spectrometer equipped with an MCT/A detector and a high-temperature reaction chamber. The samples were placed into crucible and pre-treated in pure Ar flow at 400 °C for 1 h (flow rate at 30 mL/min) and then the temperature was cooled down to 250 °C, the background data was collected at a stable state. Then, feed gases of CO (flow rate at 10 mL/min) and H₂ (flow rate at 20 mL/min) were simultaneously introduced into the crucible and spectra were recorded at intervals of one minute. After 10 minutes, the CO flow and H₂ flow were switched off, pure Ar flow (flow rate at 30 mL/min) was introduced into the crucible for 2 min, then the spectrum was recorded.

CO-desorption *in-situ* FTIR spectra were recorded using a Nicolet is50 FT-IR spectrometer equipped with an MCT/A detector and a high-temperature reaction chamber. As a typical run, the sample was placed into crucible and pre-reduced in pure H₂ flow at 300 °C for 5 h (flow rate at 30 mL/min), then the temperature was cooled down to 250 °C in pure Ar flow (flow rate at 30 mL/min) to collect the background spectra at a stable state. Subsequently, feed gases of CO (flow rate at 10 mL/min) and H₂ (flow rate at 20 mL/min) were simultaneously introduced into the crucible for an *in-situ* carbonization. After that, the gas was switched to Ar flow for a while, and CO adsorption was performed by switching the Ar flow to a dry CO flow (flow rate at 30 mL/min) at 250 °C for 0.5 h. After the CO adsorption, the system was purged with Ar (flow rate at 10 mL/min) until the gas phase CO was removed. For studying the hindered CO adsorption by water on the catalyst surface, the FTIR study was followed by similar procedures mentioned above except using the humid CO feed, which was realized by water bubbling at 20 °C.

CO adsorption in pulsed experiment was performed using a fixed-bed glass reactor connected to a mass spectrometer (SRD200M, TILON). As a typical run, 0.2 g of CoMnC (or the power mixture of 0.2 g of CoMnC and 0.2 g of PDVB) was localized in the middle of the reactor tube (40-60 mesh). The temperature of catalyst bed was raised to 250 °C in a pure He flow (30 mL/min) and maintain at 250 °C for 1 h. After that, a slight amount of CO (1 mL of mixed gas containing 10% CO/He) was pulsed into the system, the signals of CO were collected by a mass detector. The pulsed experiment was repeated for multiple times (time interval of

CO pulse was ~3 min) in each test. The CO adsorption in pulsed experiment with water injection was performed via the similar procedures except using 10% CO/He flow with water bubbling. The CO adsorption in olefin atmosphere was performed via the similar procedures except using 10% C₃H₆/He was flow gas instead of pure He.

Propylene breakthrough experiments were performed using a stainless-steel column (4.6 mm inner diameter × 50 mm). The column packed with sample was firstly purged with N₂ flow (30 mL/min) for 24 h at room temperature. After that, 10% C₃H₆/N₂ was introduced with flow rate at 10 mL/min. The propylene concentration in outlet gas was monitored by a gas chromatography (FuLi-9790, FID) equipped with a capillary PLOT-Al₂O₃ column (50 m × 0.53 mm × 0.25 μm).

Temperature-programmed surface reaction (TPSR) was performed using a fixed-bed glass reactor connected to a mass spectrometer (SRD200M, TILON). 0.2 g of CoMnC (or the power mixture of 0.2 g of steady state CoMnC and 0.2 g of PDVB) was localized in the middle of the reactor tube (40-60 mesh) and pre-treated in pure He flow at 150 °C for 1 h (flow rate at 20 mL/min). Then, the temperature was cooled to 45 °C. TPSR profiles (*m/z* at 16, 18, and 42) were collected in flowing syngas (5 mL/min, H₂/CO molar rate at 2) by raising the temperature from 45 to 300 °C with a ramping rate at 10 °C/min. The thermal stability of PDVB at the reaction temperature was evaluated using a chemisorption instrument (iChem700, PhysiChem Instruments Ltd.) connected to a mass spectrometer (SRD200M, TILON).

Catalyst performance

The syngas conversion was performed in a stainless steel fixed-bed flow reactor with an inner diameter of 10 mm. Typically, 2.0 g of composited catalyst (40-60 mesh) with weight ratio of CoMn/promoter at 1 was mixed and localized in the middle of the reactor unless otherwise stated. Additional quartz sands were placed into the reactor to maintain the catalyst bed. Prior to reaction tests, the catalyst was *in-situ* reduced at 300 °C for 5 h in a pure H₂ flow (flow rate at 20 mL/min) at atmospheric pressure. After reduction, the reactor was cooled to 250 °C, and the feed gas (H₂/CO molar ratio at 2) was introduced into the reactor under a pressure of 0.1 MPa. The gaseous products from the reactor were analyzed with two online gas chromatographs (FuLi-9790 gas chromatograph). One is equipped with a TDX-01 column (3 m × 3 mm) and a thermal conductivity detector (TCD), and the other is equipped with a capillary PLOT-Al₂O₃ column (50 m × 0.53 mm × 0.25 μm) and a flame ionization detector (FID). Argon was used as an internal standard for calculating CO conversion and product selectivities (CO₂ and CH₄) in TCD. CH₄ was used as an internal standard for calculating the product selectivities in online FID. The conversions and product selectivities were calculated according to the number

of carbon atoms. The GHSV was given on the basis of the weight of CoMnC in the reactor ($\text{mL} \cdot \text{h}^{-1} \cdot \text{g}_{\text{CoMnC}}^{-1}$), and the promoter was not included.

Computational Methods

Computational Models. Both periodic model and surface model were used for studying the diffusion behaviors of water molecules in simplified hydrophilic and hydrophobic system. For periodic models, the simulation box was $45.5 \times 59.0 \times 32.4 \text{ \AA}$, and 50 H_2O molecules adsorption in the $20 \text{ \AA}$ vacuum layer parallels to the surface (Figs. 4A and 4B). While for surface models, the simulation box was $540.4 \times 39.7 \times 32.4 \text{ \AA}$, and 100 H_2O molecules adsorption in $20 \text{ \AA}$ vacuum layer parallels to the surface. The $-\text{OH}$ and $-\text{CH}_3$ groups were added to SiO_2 (quartz) to represent hydrophilic and hydrophobic surface, respectively. The influence of water concentration on diffusion rate was also simulated by regulating the number of water molecules adsorbed on the model surface (e.g. water molecules at 25, 50, and 100, donated at 25 H_2O model, 50 H_2O model, and 100 H_2O model).

Molecular dynamics (MD) simulation. MD simulations were performed in the canonical ensemble (NVT), where the number of particles (N), volume (V), and temperature (T) were kept constant. The simulated temperature was $250 \text{ }^\circ\text{C}$ and controlled by a Nosé-Hoover thermostat with a coupling time constant of 0.1 ps . The leapfrog Verlet algorithm was used to integrate the Newton's equations of motion with a time step of 0.5 fs . 1.0×10^6 (large surface model) to 1×10^7 (periodic model) time steps were production, and the trajectories were recorded every 1000 steps for studying the diffusion behaviors of water molecules. The COMPASS (38-41) force field was used and the cutoff radius was $12.5 \text{ \AA}$. During the simulation, the Si atoms were fixed and 5 independent MD simulations were carried out for better statistics. All simulations were performed in Computing Center of Innovation Academy for Precision Measurement Science and Technology, CAS and Shenzhen Cloud Computing Center.

Mean square displacement (MSD) and diffusion coefficient (D_s). The MSD of water was defined as equation in the following:

$$MSD(\tau) = \frac{1}{N_m} \sum_i \frac{1}{N_\tau} \sum_{t_0}^{N_\tau} \left[r_i(t_0 + \tau) - r_i(t_0) \right]^2 \quad (1)$$

where N_m was the number of gas molecules, N_τ was the number of time origins used in calculating the average, and r_i was the coordinate of the i -th molecule. In addition, the slope of the MSD as a function of time determined the self-diffusion coefficient, D_s , which was defined according to the so-called Einstein relation.

$$MSD(\tau) = 2nD_s\tau + b \quad (2)$$

where n was the dimension of materials. In this work, $n = 2$ for the diffusion along with the surface. The reported MSD and D_s values were calculated as the average of 5 independent MD trajectories.

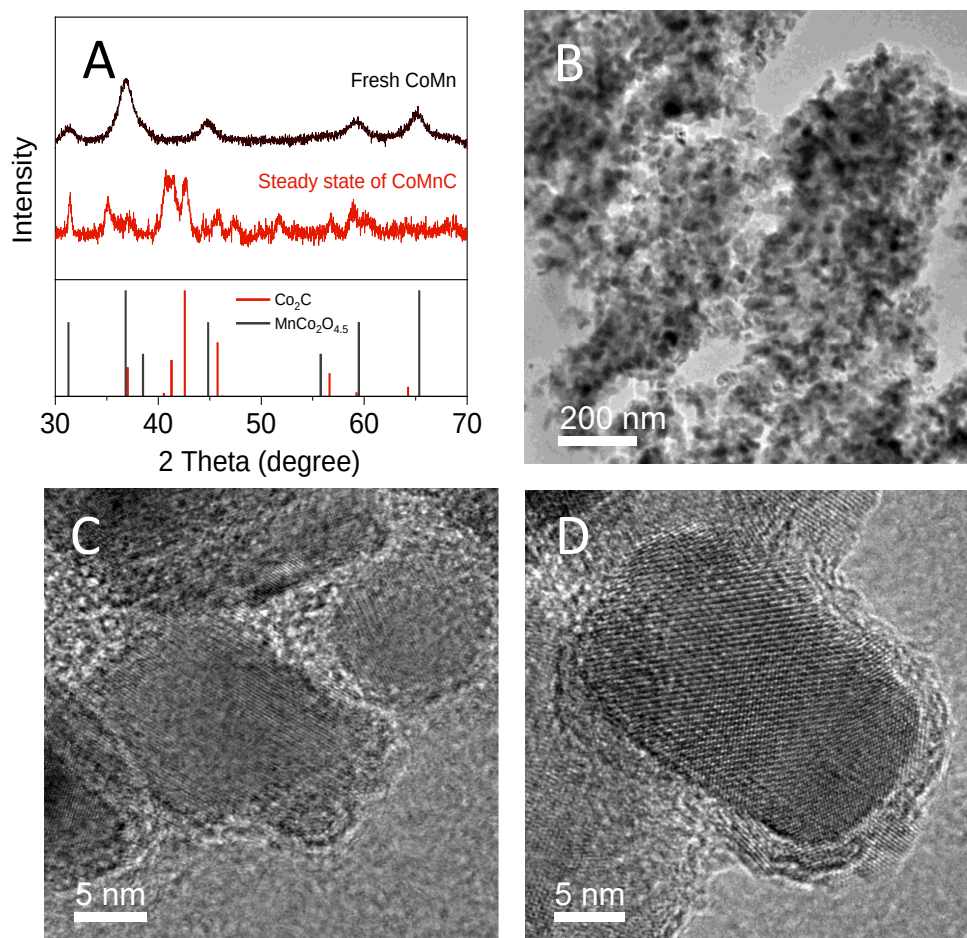


Figure S1. (A) XRD patterns of the as-synthesized CoMn oxide and steady-state CoMnC after catalyzing syngas conversion for 20 h. (B-D) TEM images of the CoMnC catalyst. Inset in (A) showed the standard XRD patterns of the Co_2C and $\text{MnCo}_2\text{O}_{4.5}$ according to PDF # 65-1457 and PDF # 32-0297, respectively.

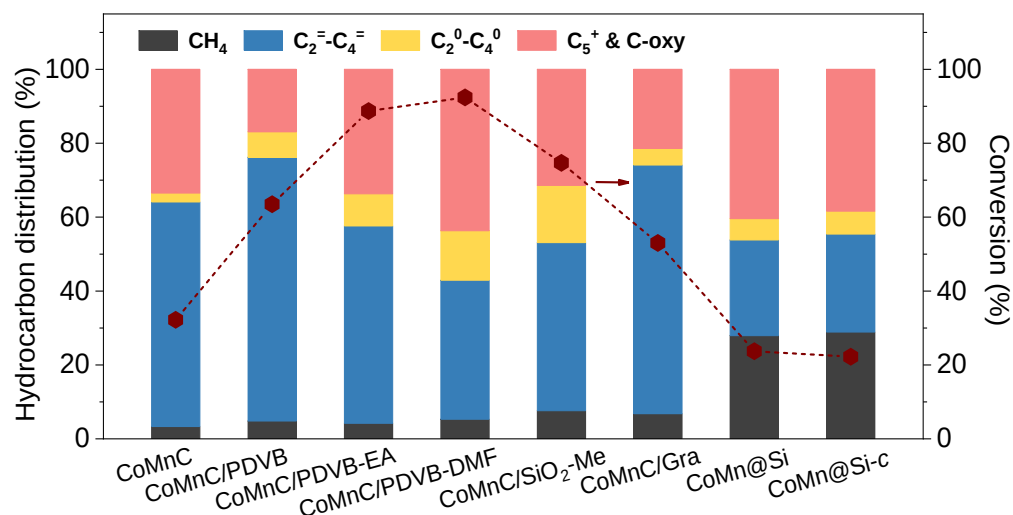


Figure S2. Catalytic data of various catalysts in syngas conversion. Reaction conditions: 1.0 g of CoMnC or 1.0 g of CoMnC physically mixed with 1.0 g of DPVB, or 1.0 g of silica modified CoMn, H₂/CO at 2, 1800 mL·h⁻¹·g⁻¹, 0.1 MPa, 250 °C, powder-mixing for CoMnC and promoter. C-oxy means the carbon-containing oxygenates.

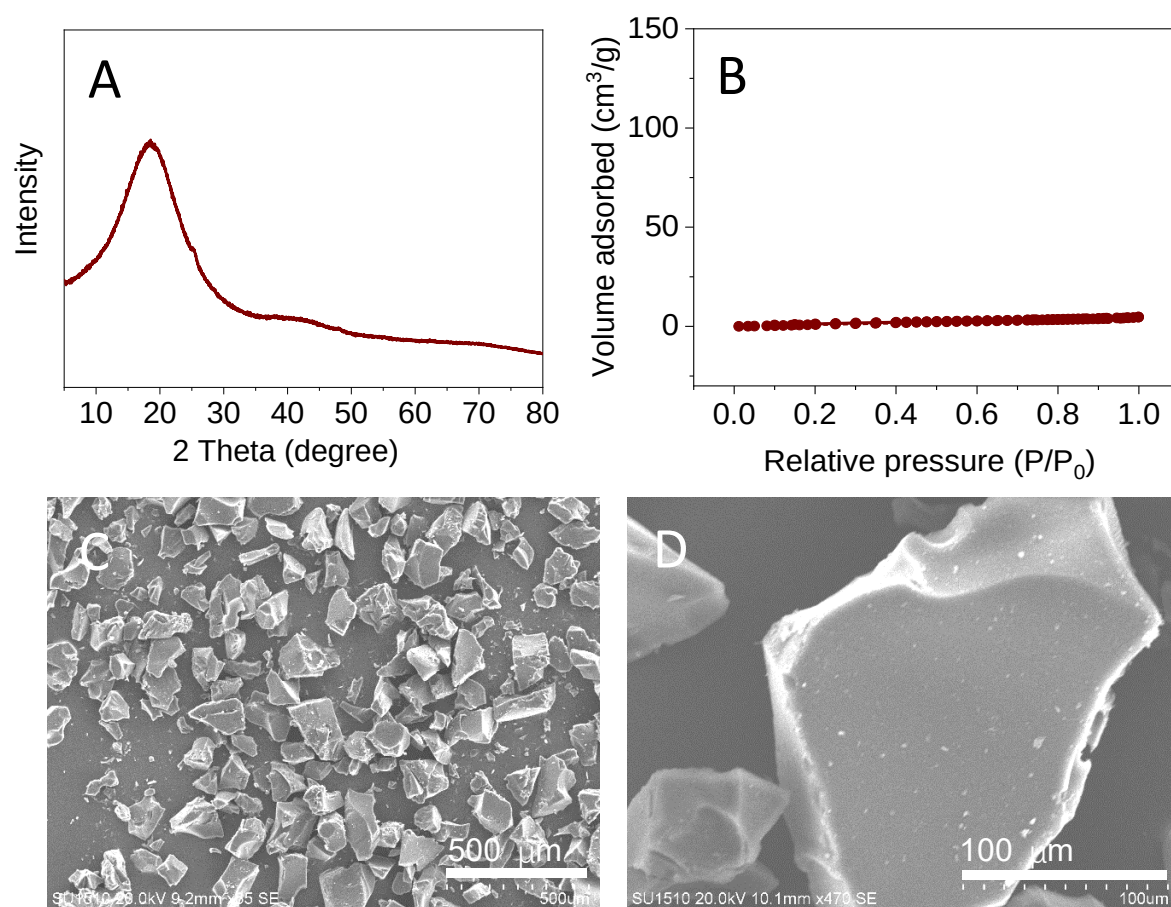


Figure S3. (A) XRD pattern, (B) N₂ sorption isotherms, and (C,D) SEM images of the PDVB.

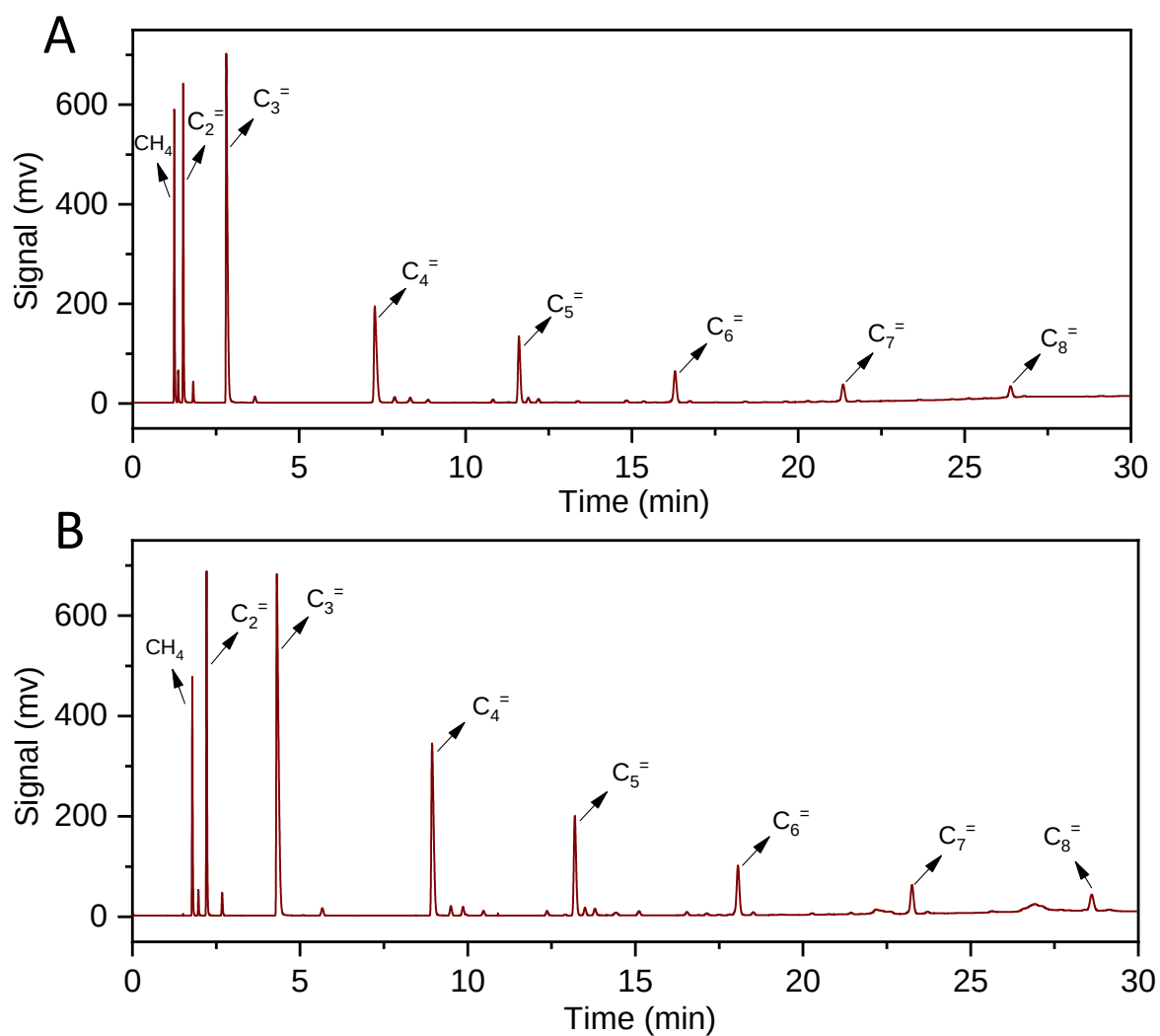


Figure S4. GC curves analyzing the gaseous products (online analysis) in the reaction over the (A) CoMnC and (B) CoMnC/PDVB catalysts.

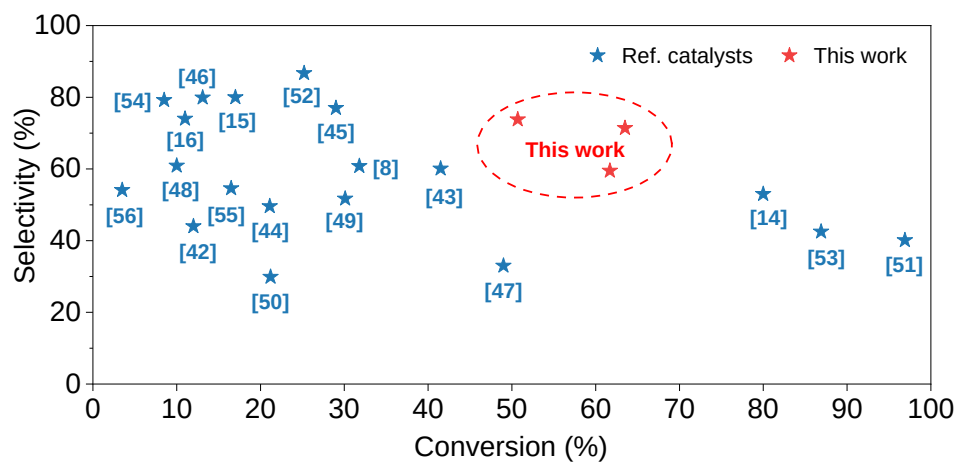


Figure S5. Selectivity of light olefins in the hydrocarbon products as a function of CO conversion over efficient catalysts in literatures and in this work. The CO₂ by-product was not included in the light olefin selectivity.

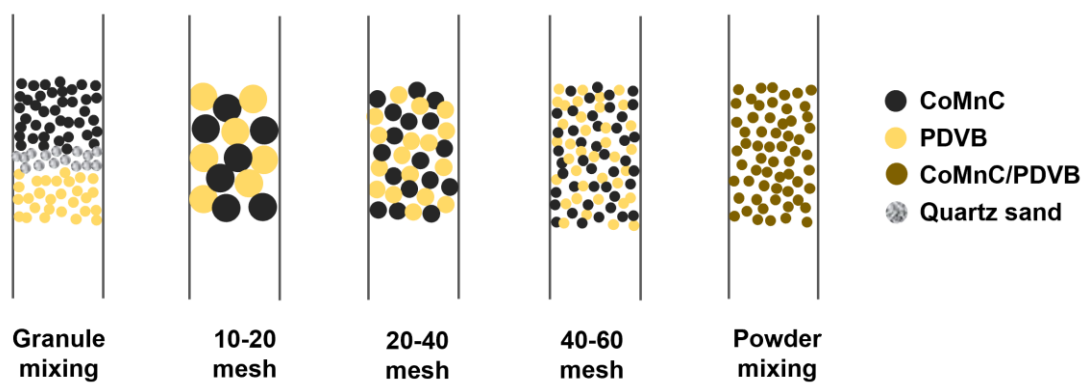


Figure S6. Modes of the CoMnC/PDVB catalysts with different proximities between CoMnC and PDVB.

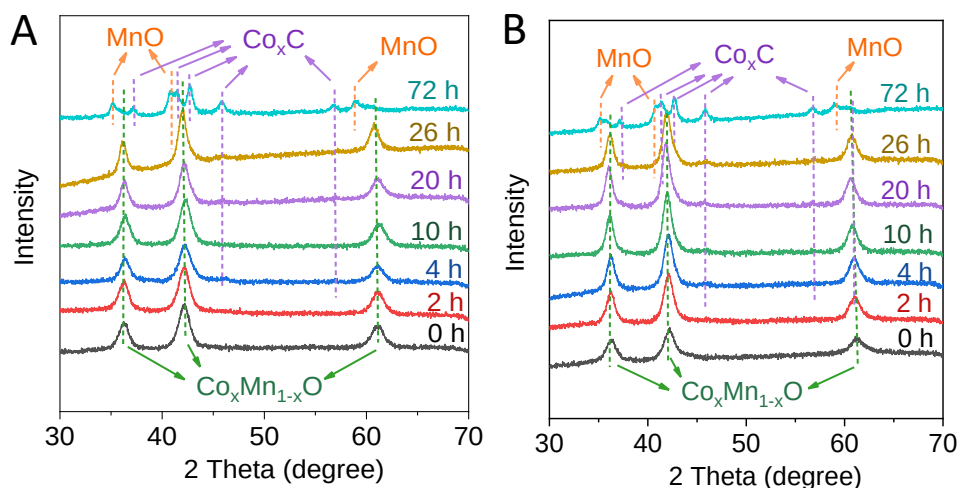


Figure S7. XRD patterns of the (A) CoMnC and (B) CoMnC/PDVB catalysts at different reaction periods. Reaction conditions: 0.2 g of CoMnC or 0.2 g of CoMnC physically mixed with 0.2 g of PDVB, H₂/CO at 2, 1800 mL·h⁻¹·g⁻¹, 0.1 MPa, 250 °C, powder-mixing for CoMnC and PDVB.

Note: In the initial 15 h of the syngas conversion (Fig. 2C), the CO conversion was continuously increased from 10.2% to 67.6% in this period and became constant after 15 h. The selectivity to C₂-C₄ olefins was also increased from 18.1% to 68.2%, accompanied with the decrease of C₅₊ product selectivity. Such reaction trend is very similar to that in the syngas conversion on general CoMnC catalyst without PDVB, where the conversion/selectivity changed with the CoMnC carbonization in the initial reaction (8).

In addition, this hypothesis on PDVB (or the depolymerized species) participating in the CoMnC carbonization could be excluded by characterizing the CoMnC phase change as a function of reaction time. Negligible difference was observed in the XRD patterns of the CoMnC with and without PDVB during the initial reaction periods (fig. S7), as well as the TEM characterizations (fig. S8). These data further support that the PDVB in physically mixed manner does not change the catalyst structure.

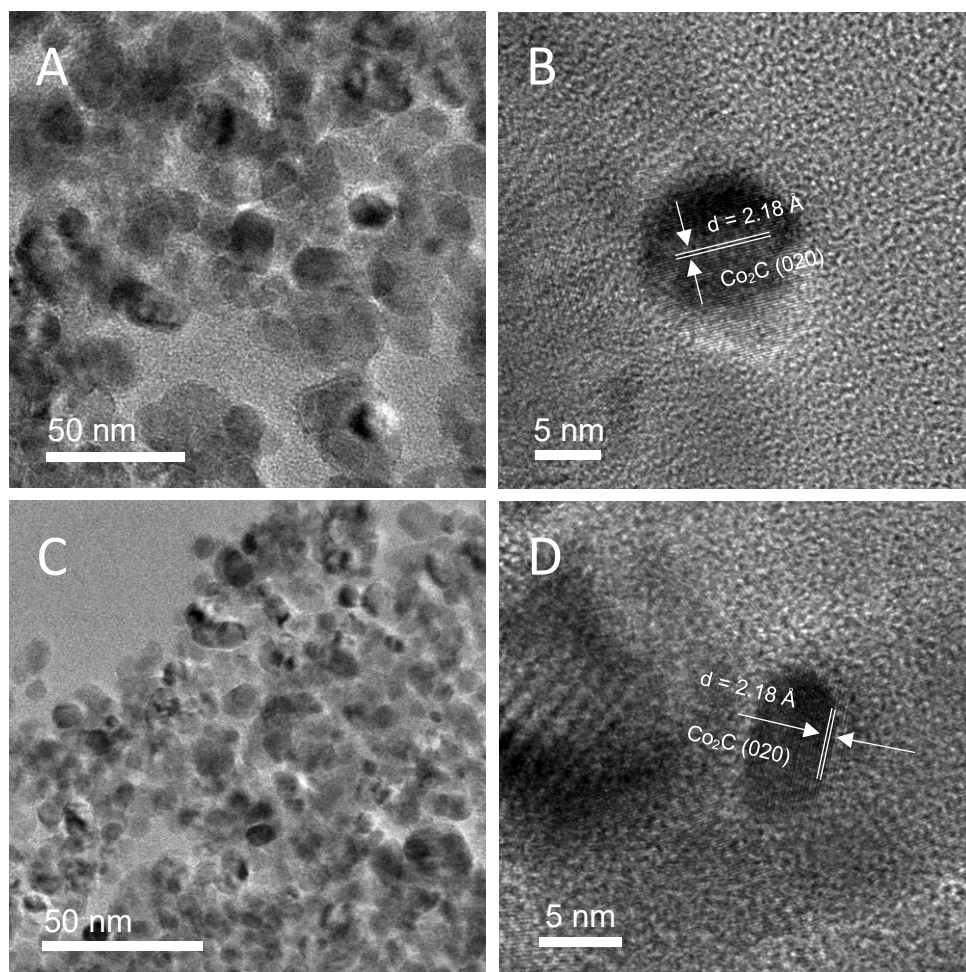


Figure S8. TEM images of the (A,B) CoMnC catalyst and (C,D) CoMnC separated from CoMnC/PDVB catalyst after catalyzing the syngas conversion for 72 h.

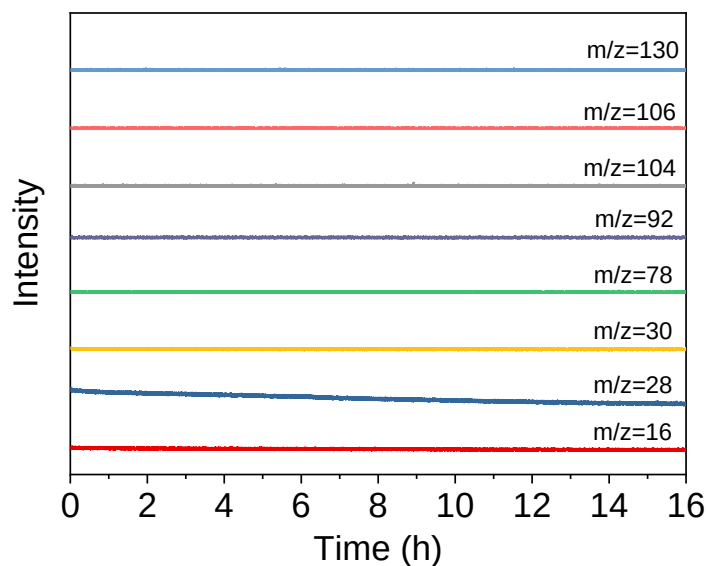


Figure S9. Mass spectra characterizing PDVB stability at 250 °C for more than 16 h.

Note: To evaluate the stability of PDVB, the PDVB was heated in a nitrogen flow at 250 °C and the emission gas was analyzed by a mass spectroscopy. As shown in fig. S8, the signals of possible species from PDVB decomposition (m/z at 16, 28, 30, 78, 92, 104, 106, and 130) were undetectable in the mass spectra. The ^{13}C NMR spectra of PDVB were unchanged before and after the thermal treatment at 250 °C for 10 h (fig. S10). Fig. S11A shows the TG-MS profiles of PDVB, where the weight loss starts at ~ 370 °C with the formation of multiple hydrocarbon species detected by mass spectrometer, suggesting its decomposition at the temperature higher than 370 °C. At 250 °C, the reaction temperature for syngas conversion, the weight loss and decomposed species were completely undetectable. These data confirm the good stability of PDVB under hours of annealing at the reaction temperature. Fig. S11B showed the TG-DSC profiles of PDVB, where the endothermal signal started at about 300 °C, which suggest that the PDVB might melt at temperatures higher than 300 °C but very stable at 250 °C. In addition, we also heated the PDVB granules in a quartz tube at 250 °C for a long period of 50 h, it was observed that the morphology of the granules was unchanged, suggesting the excellent physical stability (fig. S12). This feature might help to exclude the possibility of PDVB melting and coating on the CoMnN surface to change the performances.

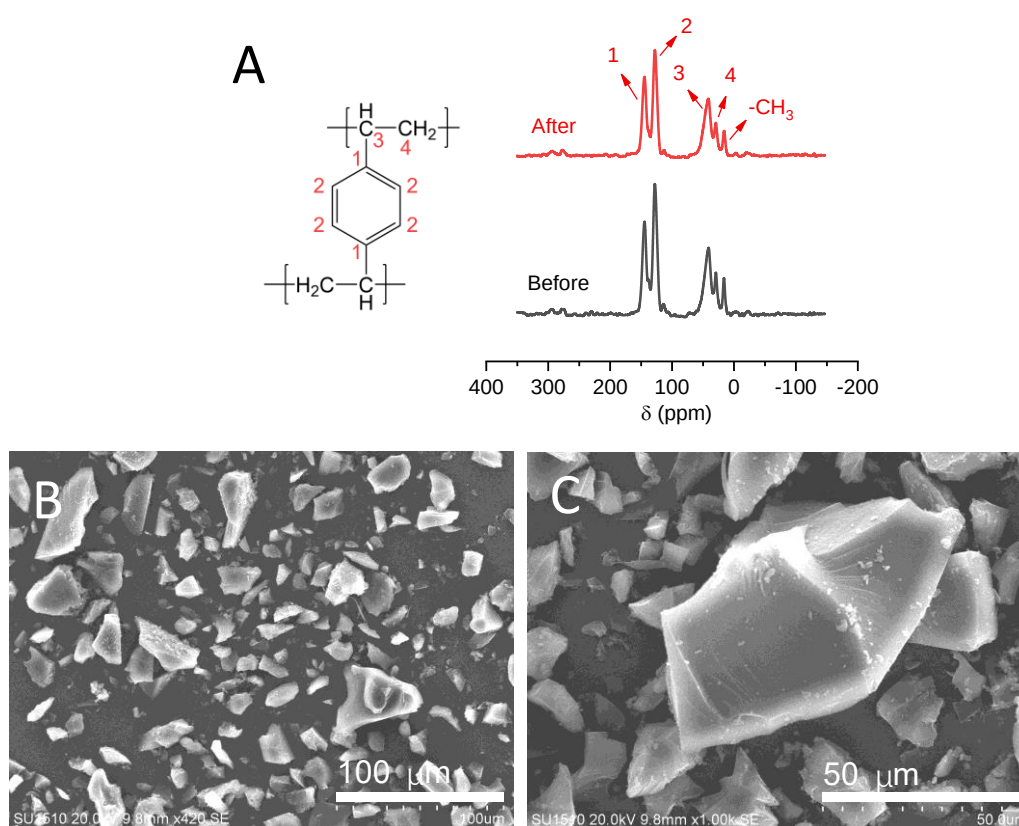


Figure S10. (A) ^{13}C NMR spectra of PDVB before and after treatment at 250 °C for 10 h. (B,C) SEM images of PDVB separated from CoMnC/PDVB catalyst after reaction.

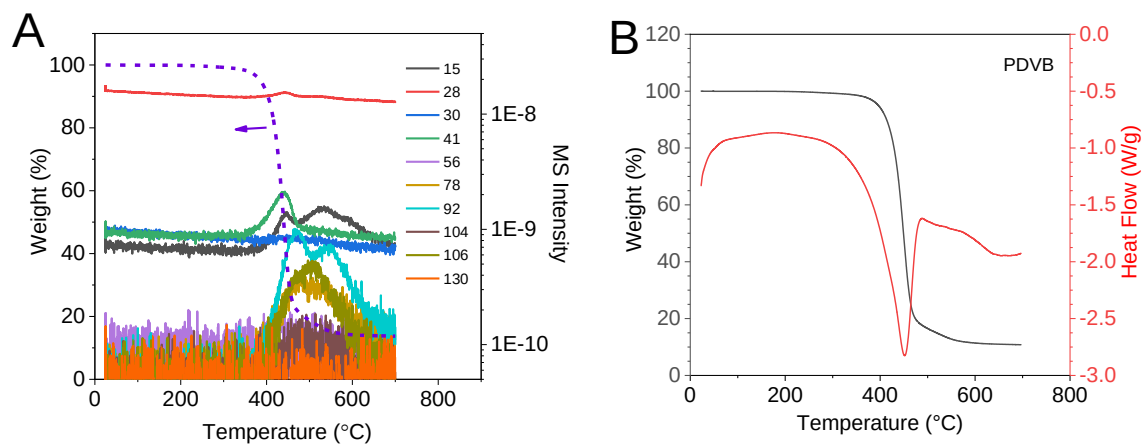


Figure S11. (A) TG-MS and (B) TG-DSC profiles of the PDVB.

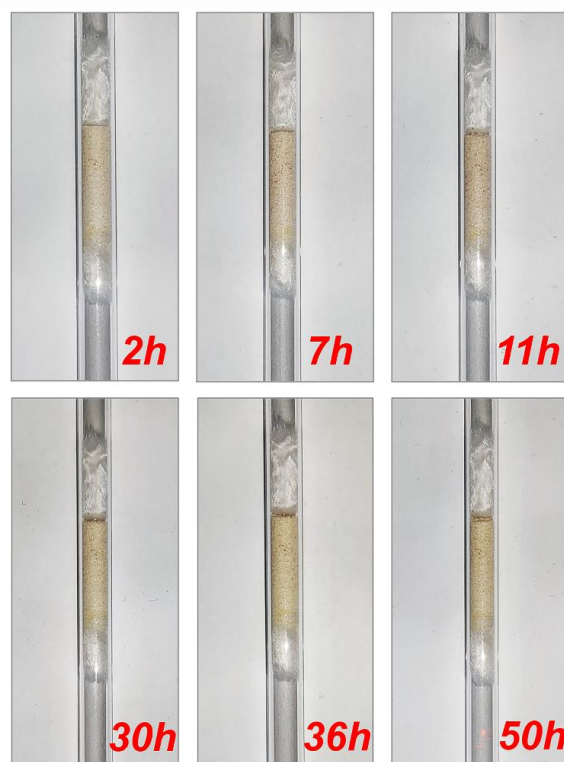


Figure S12. Photographs showing the PDVB granules in a quartz tube with thermal treatment at 250 °C for different periods.

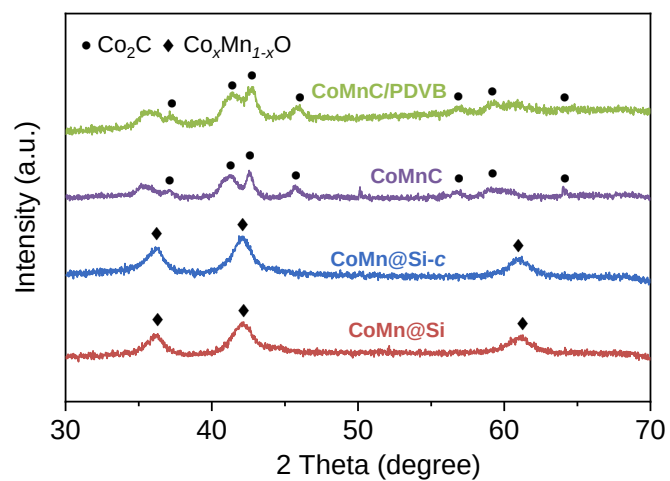


Figure S13. XRD patterns of the used CoMn@Si, CoMn@Si-c, CoMnC, and CoMnC/PDVB catalysts.

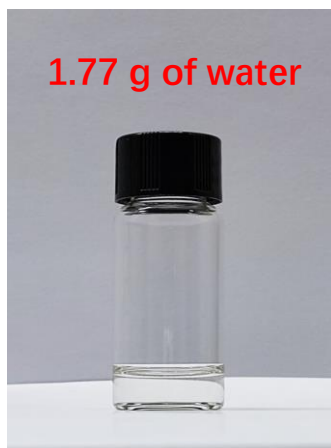


Figure S14. Photograph showing water in the final products from CoMnC/PDVB catalyzed syngas conversion. This water was collected for 102 h.

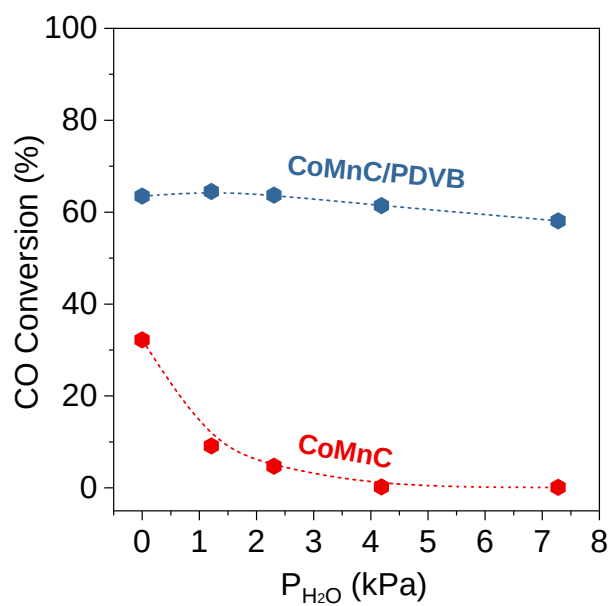


Figure S15. Dependences of CO conversions on water concentrations over the CoMnC and CoMnC/PDVB catalysts. Reaction conditions: 1.0 g of CoMnC or 1.0 g of CoMnC physically mixed with 1.0 g of PDVB, H_2/CO at 2, $1800\text{ mL}\cdot\text{h}^{-1}\cdot\text{g}^{-1}$, 0.1 MPa, 250 °C, powder-mixing for CoMnC and PDVB. Water bubbling with CO flow at 10, 20, 30, and 40 °C to adjust the partial pressures. The water feed rate at $\sim 10.5\text{ mg g}_{CoMnC}^{-1}\text{ h}^{-1}$ was realized by water bubbling at 10 °C, corresponding to the partial pressure of $\sim 1.2\text{ kPa}$.

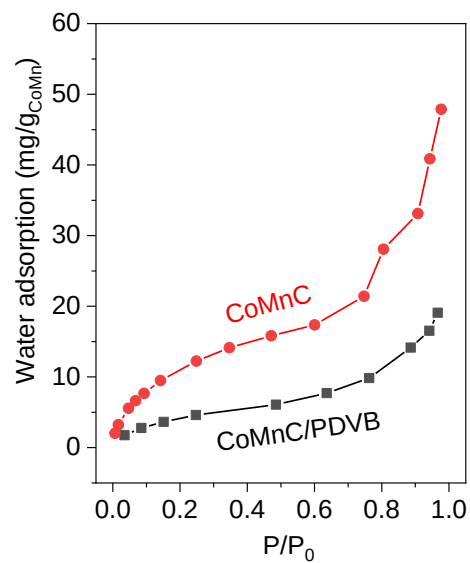


Figure S16. Water adsorption tests on the CoMnC and CoMnC/PDVB.

Note: We tested the water adsorption ability of the CoMnC and CoMnC/PDVB at different relative humidity. The CoMnC always showed higher water adsorption ability than CoMnC/PDVB at the equivalent relative pressures of water (P/P_0). These data confirm the efficiently hindered water adsorption on the CoMnC by mixing with hydrophobic PDVB.

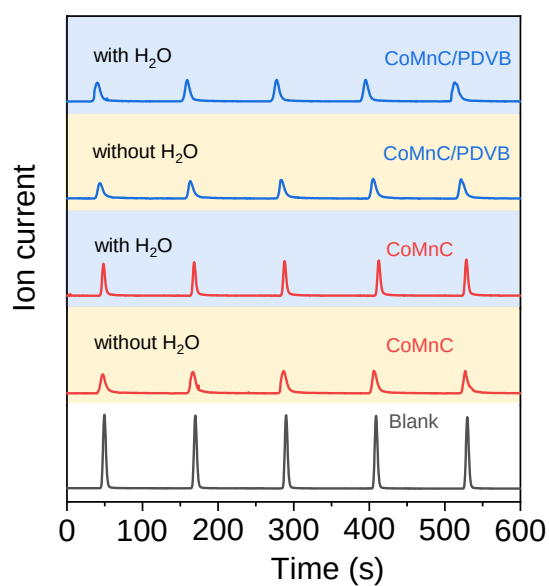


Figure S17. Transient response curves for pulsed CO (5 mL/min) in Ar flow (30 mL/min He) at 250 °C over the CoMnC and CoMnC/PDVB catalysts. Water bubbling with CO flow at 10 °C (water concentration at 1.2 vol% in CO) for introduction to the system.

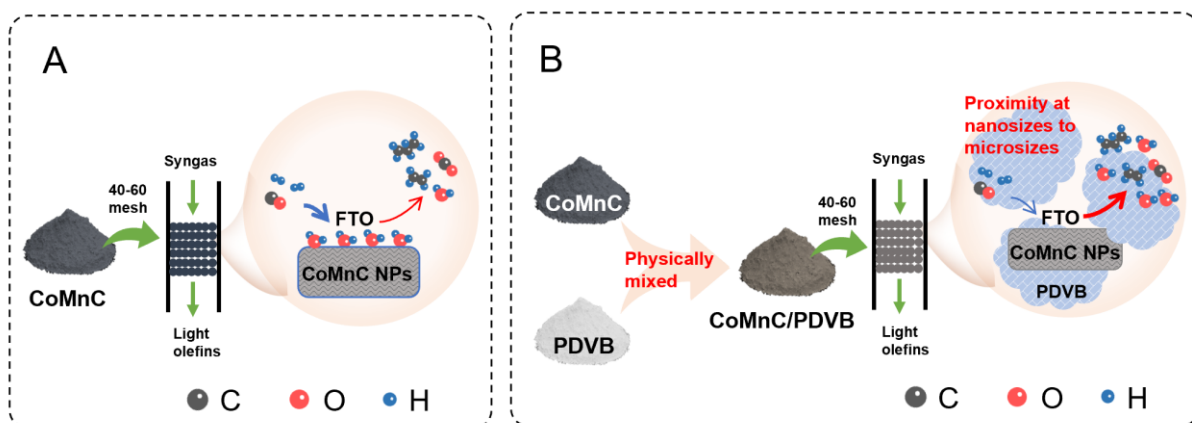


Figure S18. Scheme showing the CoMnC physically mixed with PDVB catalyst greatly boost the activity of syngas conversion *via* selectively and rapidly removing the water products from reaction system.

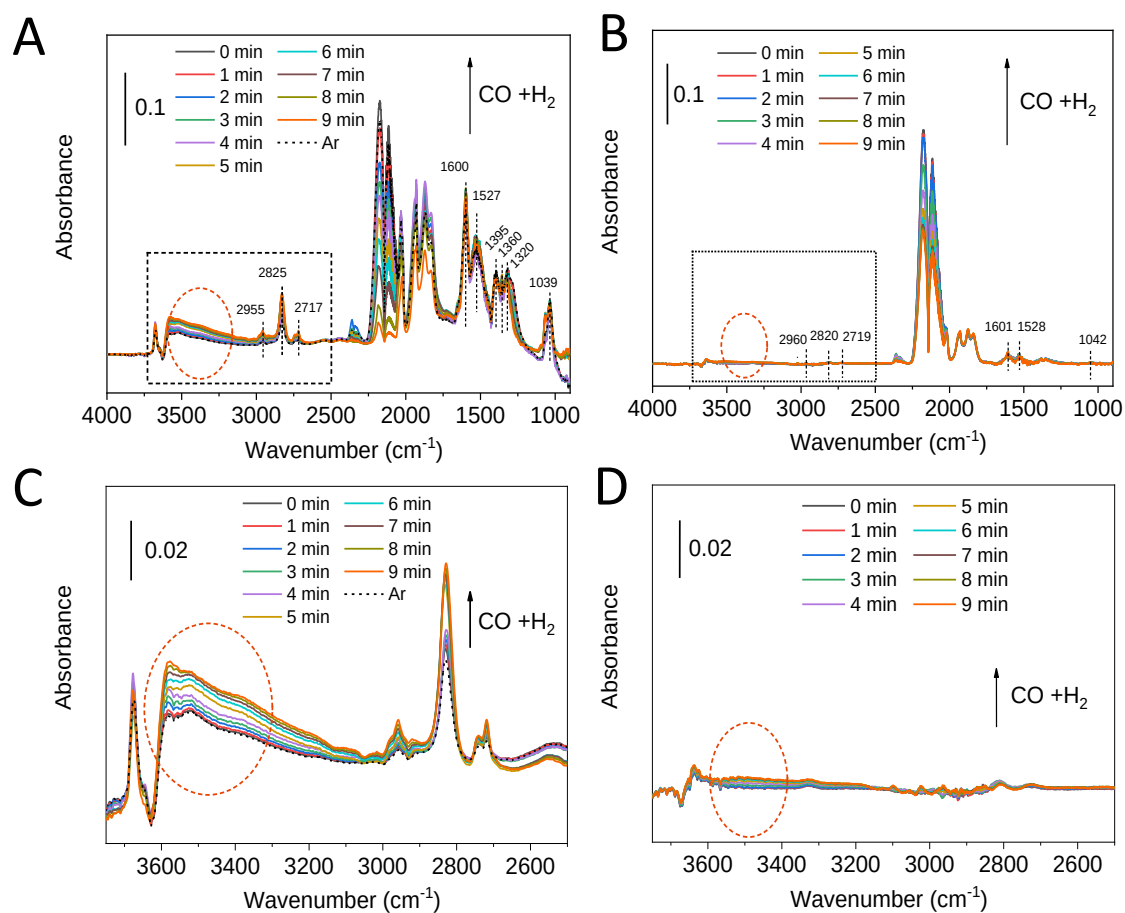


Figure S19. CO-hydrogenation *in-situ* FTIR spectra of the (A) CoMnC and (B) CoMnC/PDVB. (C) enlarge view of the region at 2500-3750 cm^{-1} in fig. S19A. (D) Enlarge view of the region at 2500-3750 cm^{-1} in fig. S19B.

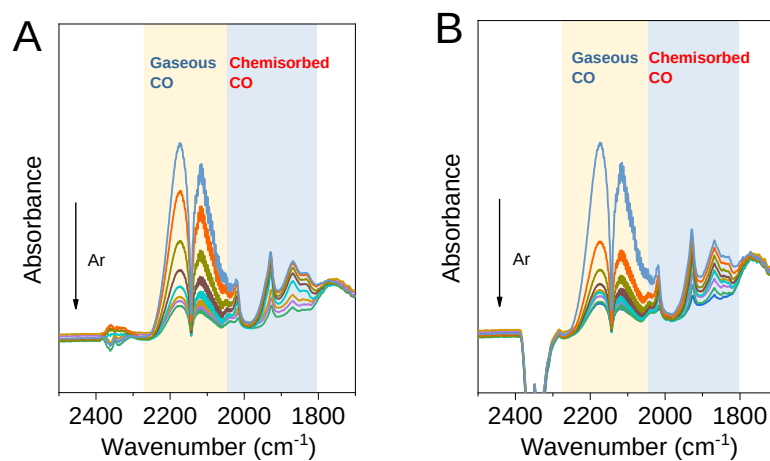


Figure S20. CO-desorption FTIR spectra of (A) anhydrous and (B) water-pretreated CoMnC/PDVB.

Note: Introducing CO to the CoMnC/PDVB catalysts showed the signals chemically adsorbed CO on the CoMnC surface in the *In-situ* FTIR spectra. By feeding water to the sample with CO gas, these signals were almost unchanged, supporting the negligible influence of water to CO adsorption on CoMnC, which is quite different from the water hindered CO adsorption on bare CoMnC without PDVB (Figs. 3D and E).

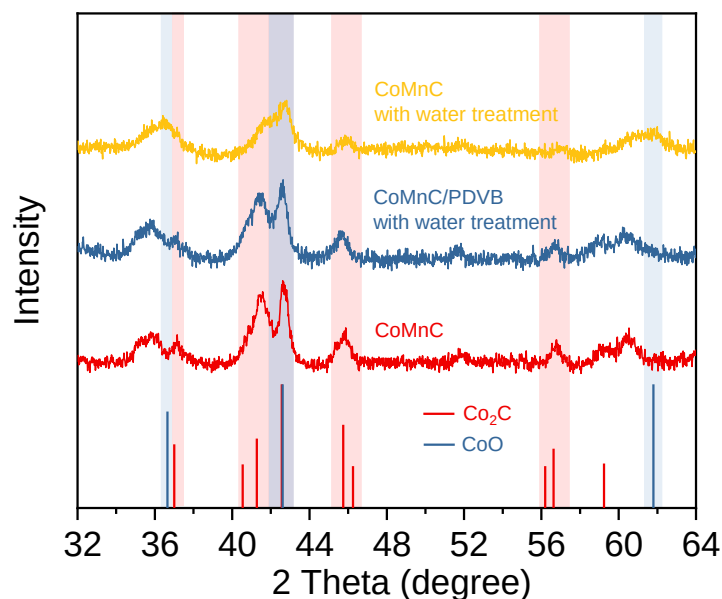


Figure S21. XRD patterns for studying the ability of the PDVB in improving the water resistance of CoMnC. The standard XRD patterns of the Co₂C and CoO is according to PDF # 65-1457 and PDF # 01-1227, respectively.

Note: The metal carbide catalysts for syngas conversion usually suffer from the oxidation by water to form the oxide phase under the long-period reaction or special treatment conditions. Therefore, we investigated the ability of the PDVB in improving the water resistance of CoMnC. Figure S21 showed that CoMnC could be oxidized by water treatment at 250 °C, as confirmed by the reduced Co₂C peaks and appearance of CoO peak in the XRD patterns. In contrast, the CoMnC with physically mixed PDVB was stable, giving almost unchanged XRD pattern before and after the treatment. This feature confirms that the PDVB could hinder the access of water to the cobalt carbide species to avoid the carbide oxidation, which helps to understand the hindered re-adsorption of water product to the CoMnC surface to benefit the CO adsorption in syngas conversion.

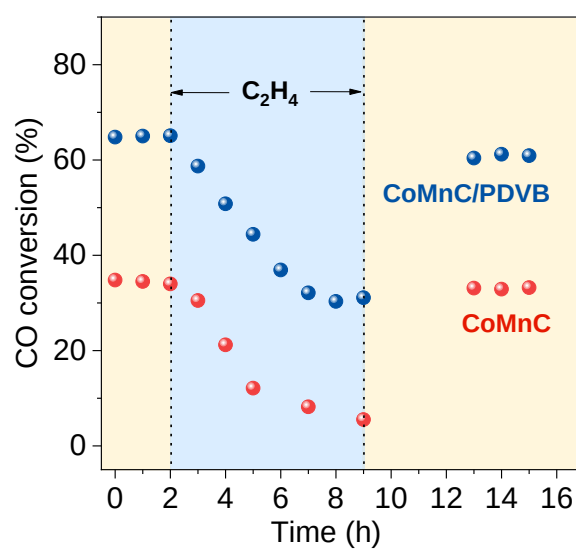


Figure S22. Data showing the influence of ethene to the syngas conversion on the CoMnC and CoMnC/PDVB catalysts. Reaction conditions: 1.0 g of CoMnC or 1.0 g of CoMnC physically mixed with 1.0 g of PDVB, ethene/H₂/CO at 0.6/2/1, 1800 mL·h⁻¹·g⁻¹, 0.1 MPa, 250 °C, powder-mixing for CoMnC and PDVB.

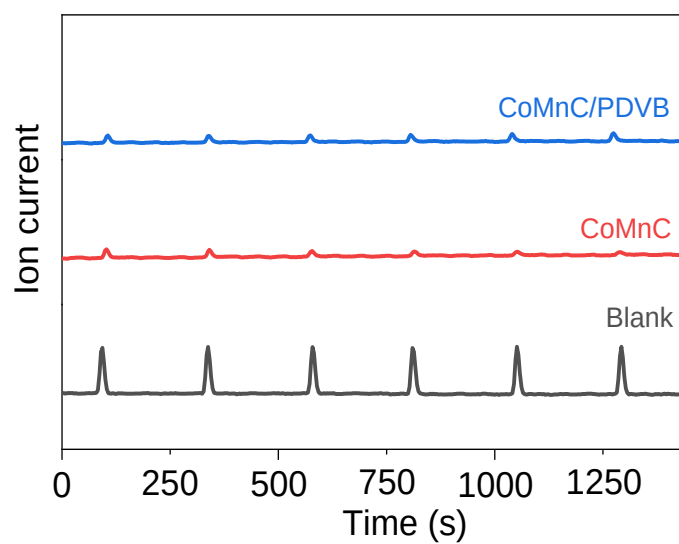


Figure S23. Transient response curves obtained during pulses of 10% CO/He (5 mL/min) into 10% C₃H₆ He flow (30mL/min He) at 250 °C over the CoMnC and CoMnC/PDVB catalysts. The CO signals (m/z at 28) were collected.

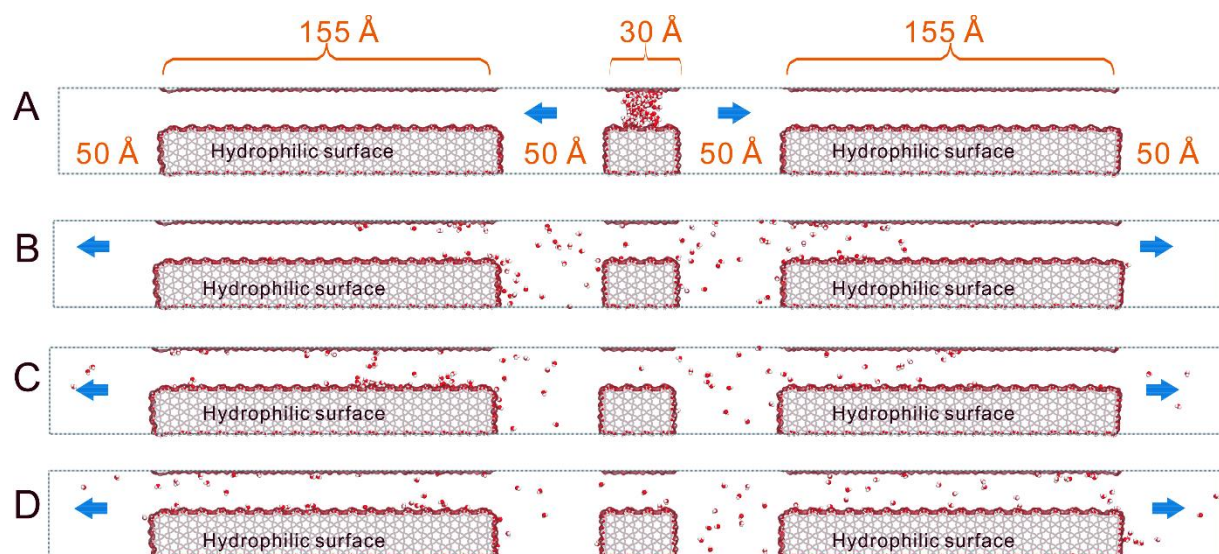


Figure S24. (A) The initial state of water molecules adsorbed on models with hydrophilic surface. Scheme showing the simulated diffusion of the water molecules on the models with hydrophilic surface at (B) 50, (C) 250, and (D) 500 ps.

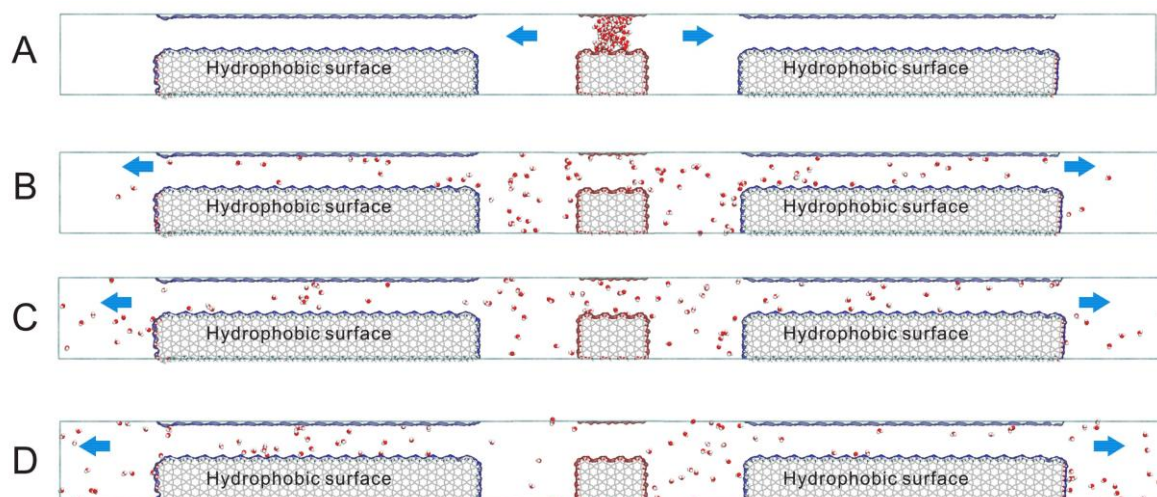


Figure S25. (A) The initial state of water molecules adsorbed on models with hydrophobic surface. Scheme showing the simulated diffusion of the water molecules on the models with hydrophobic surface at (B) 50, (C) 250, and (D) 500 ps.

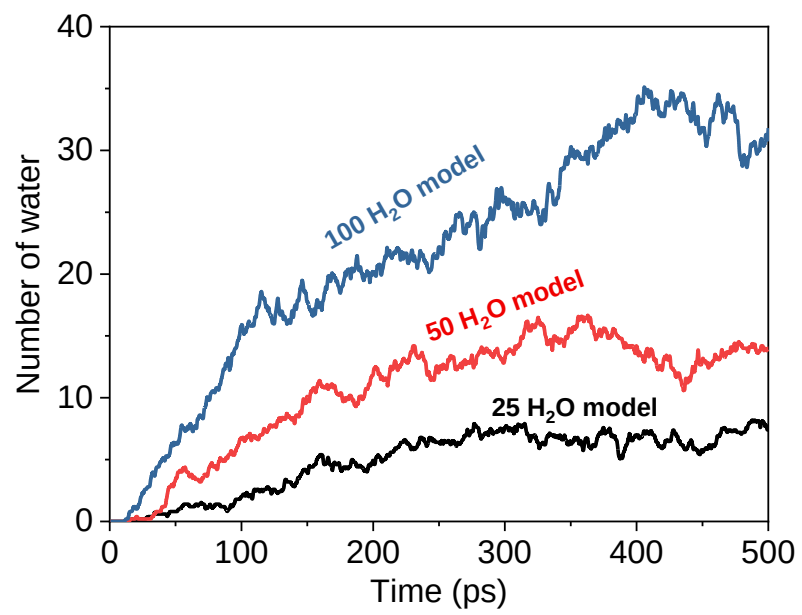
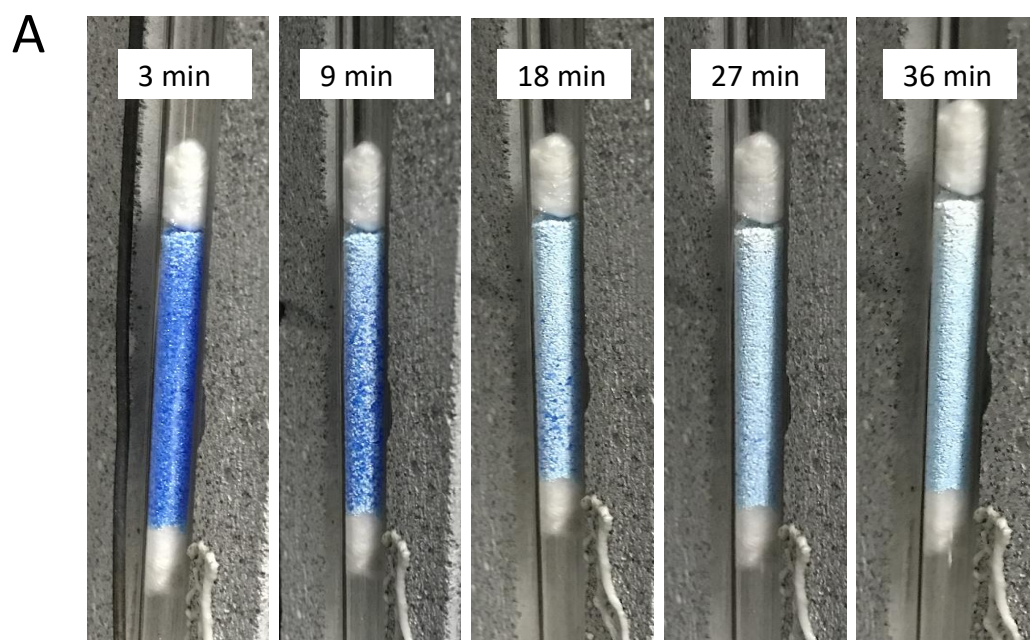
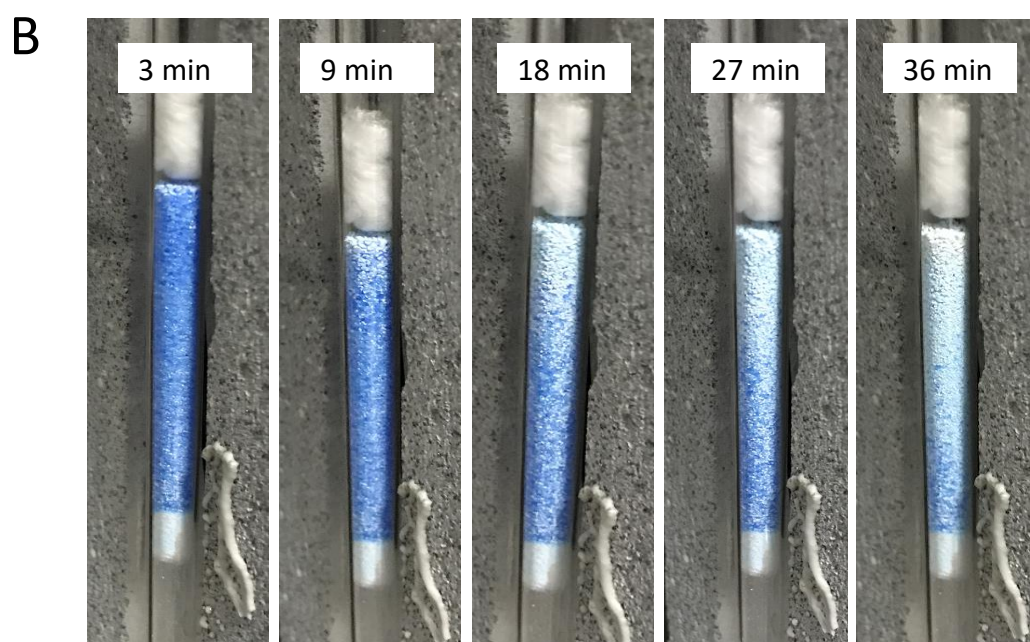


Figure S26. The number of water molecules escaped from hydrophobic model surface as a function of time.



2g $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ + 15 mg PDVB



2g $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$

Figure S27. Photographs of (A) 2 g of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ mixed with 15 mg of PDVB and (B) 2 g of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ in a N_2 flow (20 mL/min) at 100 °C for different periods.

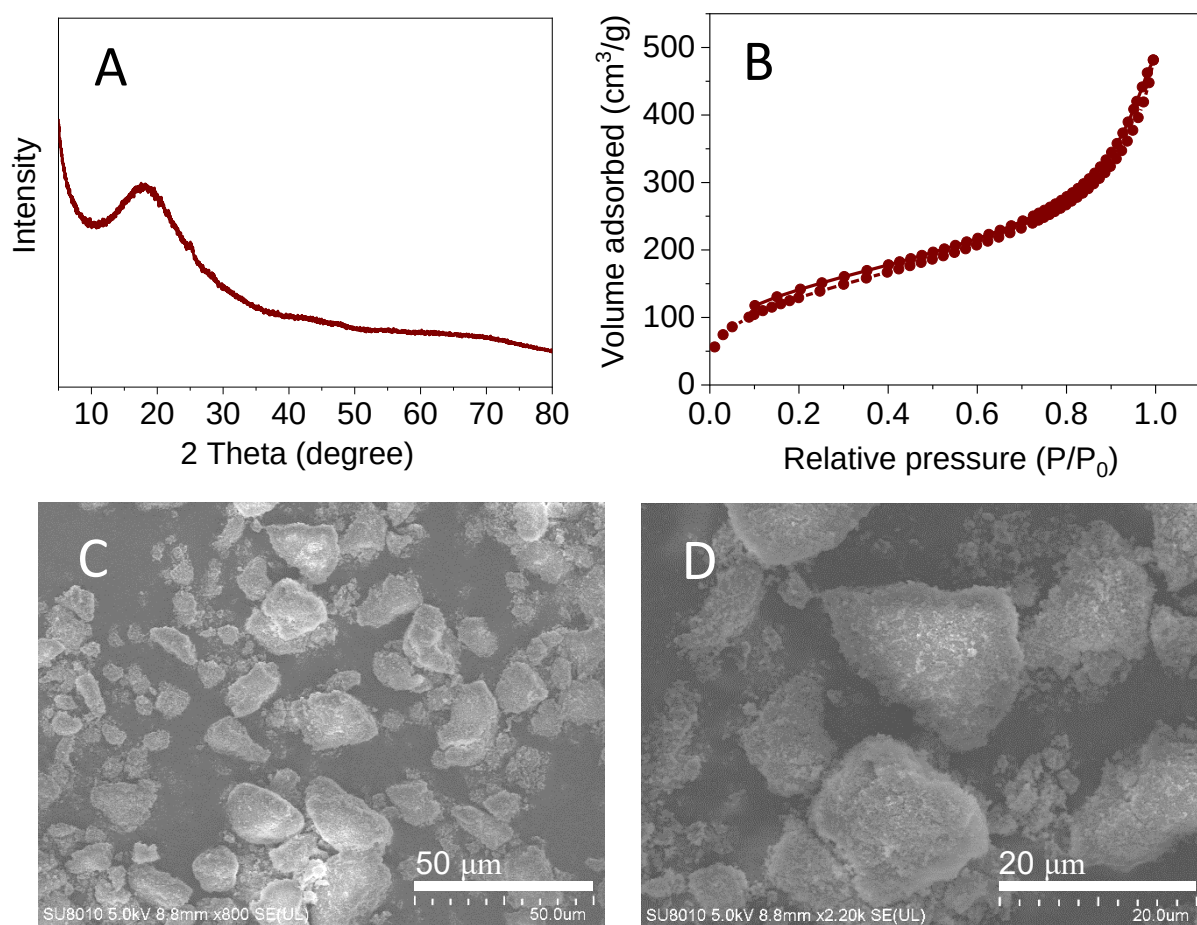


Figure S28. (A) XRD pattern, (B) N₂ sorption isotherms, and (C, D) SEM images of the PDVB-EA.

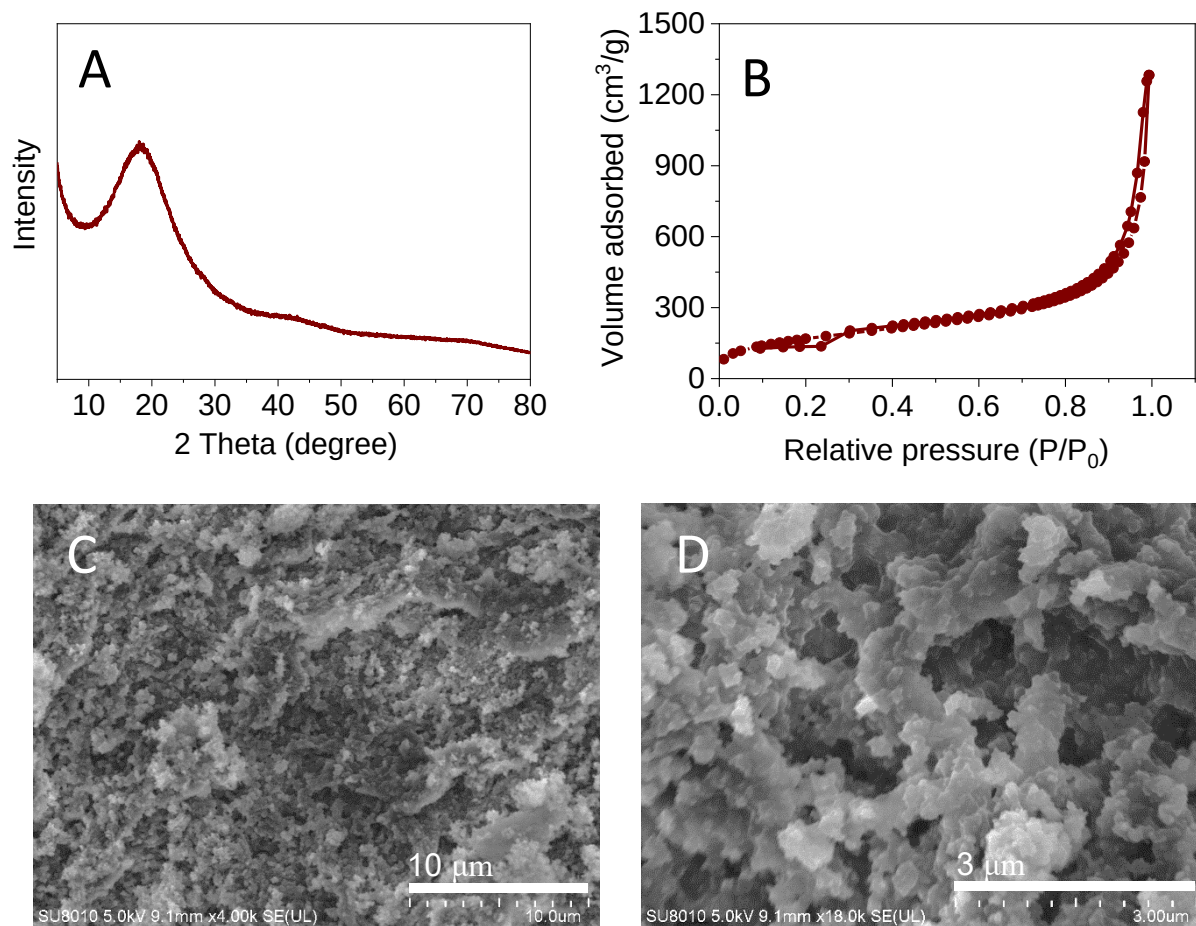


Figure S29. (A) XRD pattern, (B) N₂ sorption isotherms, and (C,D) SEM images of the PDVB-DMF.

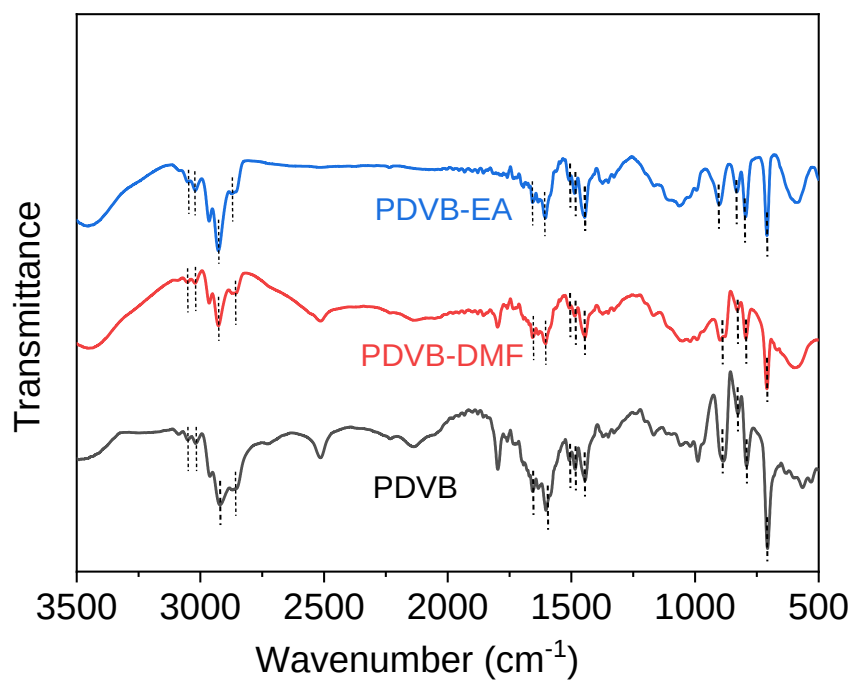


Figure S30. IR spectra of the PDVB, PDVB-DMF, and PDVB-EA.

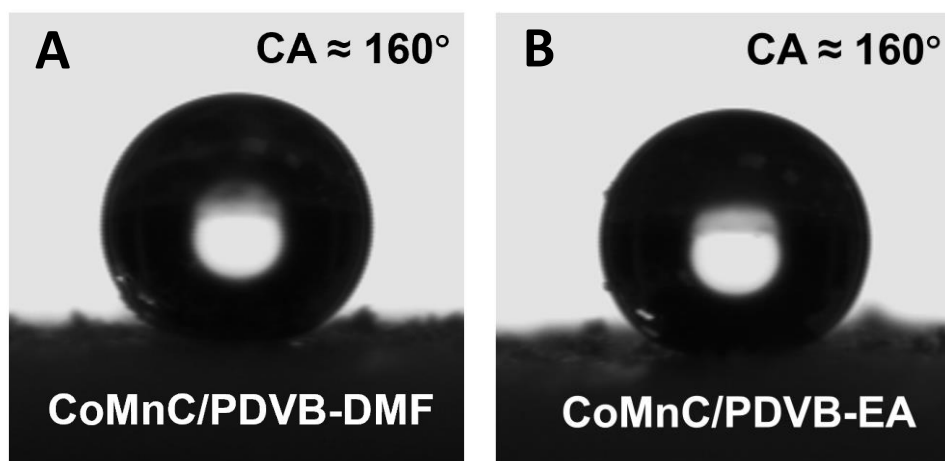


Figure S31. Water-droplet contact angles of the (A) CoMnC/PDVB-DMF and (B) CoMnC/PDVB-EA.

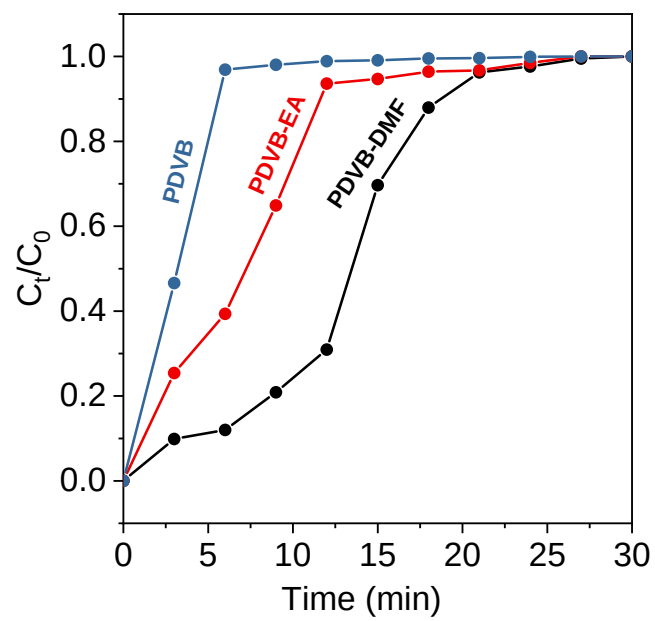


Figure S32. C_3H_6 breakthrough curves of the PDVB, PDVB-DMF, and PDVB-EA.

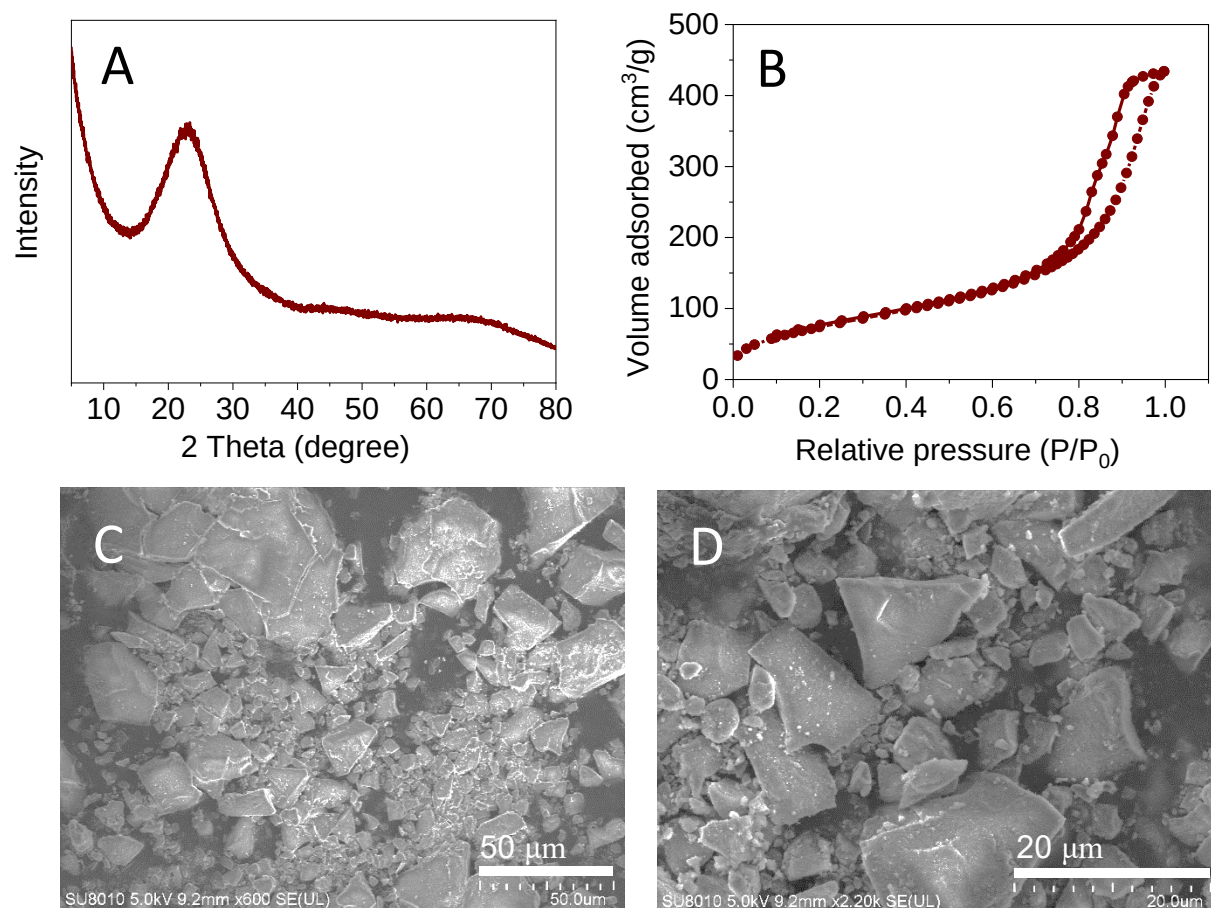


Figure S33. (A) XRD pattern, (B) N₂ sorption isotherms, and (C,D) SEM images of the SiO₂-Me.

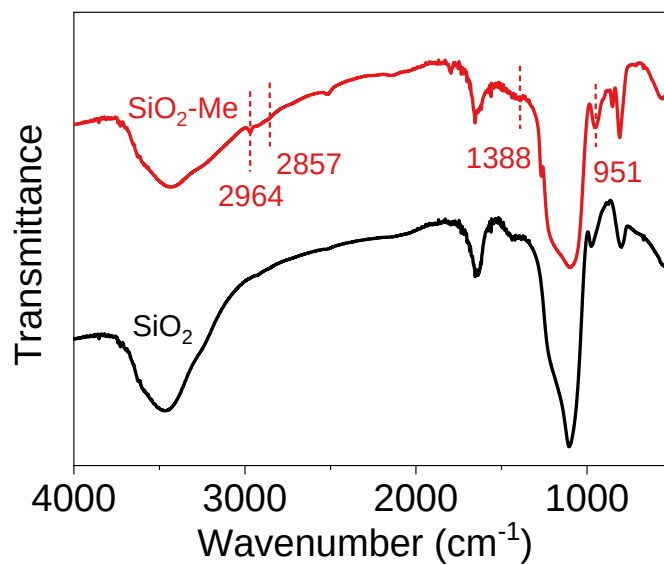


Figure S34. FTIR spectra of SiO₂ and SiO₂-Me.

Note: Compared with SiO₂, the SiO₂-Me gave additional peaks at 2964, 2857, 1388, and 951 cm⁻¹, which were assigned to the asymmetrical stretching vibration of -CH₃ (2964 cm⁻¹), symmetrical stretching vibration of -CH₃ (2857 cm⁻¹), and bending vibration of -CH₃ (1394 cm⁻¹), confirming the successful modification of methyl groups on SiO₂ (57).

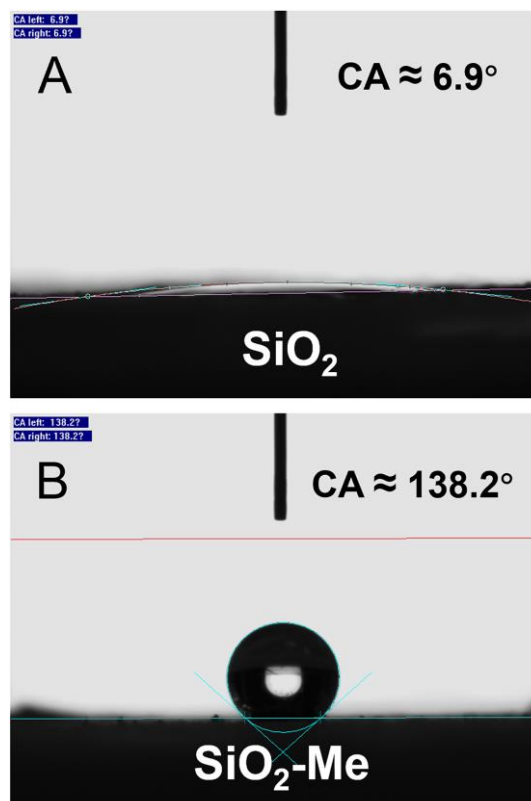


Figure S35. Water-droplet contact angles of the (A) SiO_2 and (B) $\text{SiO}_2\text{-Me}$.

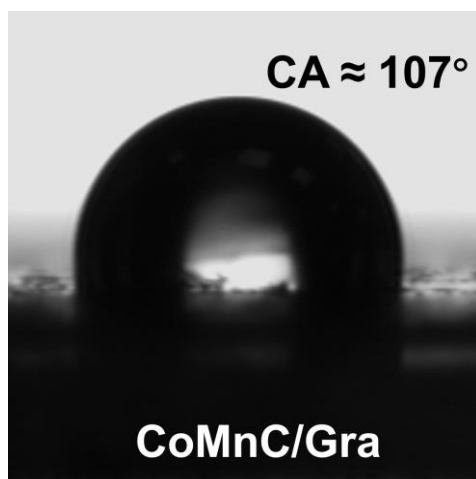


Figure S36. Water-droplet contact angle of the CoMnC/Gra catalyst.

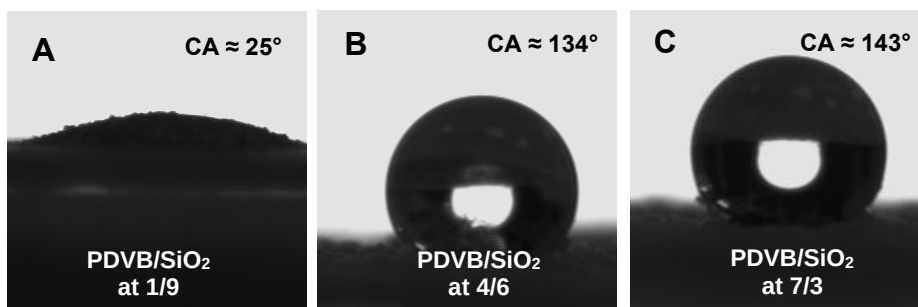


Figure S37. Water-droplet contact angles of the PDVB and SiO₂ mixtures with different PDVB/SiO₂ weight ratio at (A) 1/9, (B) 4/6, and (C) 7/3, respectively.

Note: The materials with different wettability were mixed with CoMnC and evaluated in the syngas conversion. We reasonably mixed hydrophobic PDVB and hydrophilic silica with different ratios, and the wettability of the mixtures could be varied and then used as promoters for the CoMnC catalyzed syngas conversion. Fig. S37 shows water-droplet contact angles of 25, 134, and 143° for the mixtures with different PDVB/SiO₂ weight ratios at 1/9, 4/6, and 7/3, respectively. In the syngas conversion, the CO conversions were positively related to the hydrophobicity (table S8). The conversion was 34.8% on the CoMnC catalyst containing the hydrophilic promoter (PDVB/SiO₂ weight ratio at 1/9), and increased to 52.4% and 65.9% when relatively hydrophobic PDVB/SiO₂ mixtures (PDVB/SiO₂ weight ratio at 4/6 and 7/3) were employed. In contrast, the PDVB without SiO₂ gave a CO conversion at 72.2%, supporting the importance of promoter hydrophobicity for the catalysis. In this case, more hydrophobic promoter also slightly reduced the olefin selectivity as confirmed by the changed o/p ratio in these reactions, because the rapid water escape from the catalyst surface could lead to the re-adsorption and deep hydrogenation of olefin products.

We have to note that the CoMnC and PDVB/SiO₂ mixed catalysts were made into granule mixture with sizes at 80-100 mesh in the aforementioned tests rather than the powder mixture to avoid the interaction between silanol groups with the CoMnC surface during the reaction. According to the catalysts over CoMnC/PDVB catalysts with different mixing manners in power and 80-100 mesh granules, the catalytic performances are comparable in these cases.

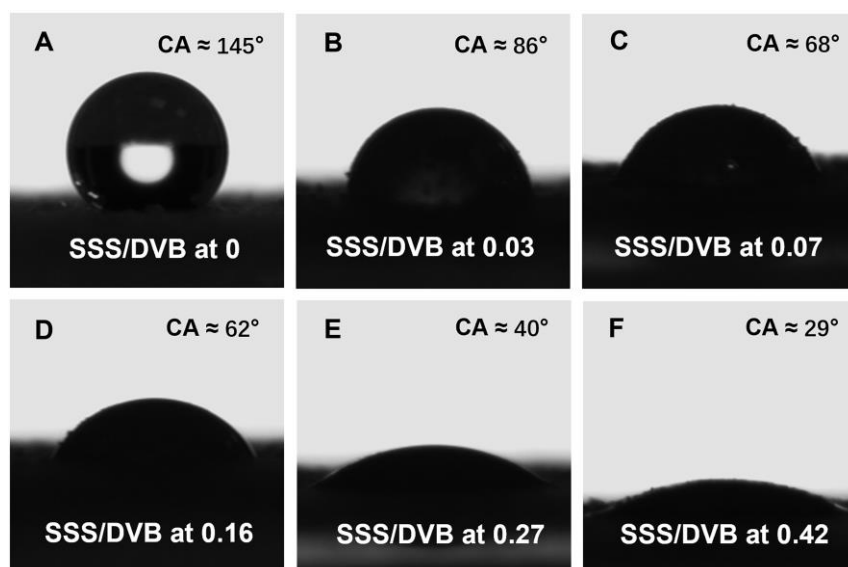


Figure S38. Water-droplet contact angles of the PDVB-SSS materials with varied molar ratios of SSS and DVB in the starting mixture at (A) 0, (B) 0.03, (C) 0.07, (D) 0.16, (E) 0.27, and (F) 0.42.

Note: We changed the wettability of the polymer support from hydrophobic to relatively hydrophilic by introducing the hydrophilic sodium styrene sulfonate (SSS) to the PDVB. By varying the molar ratios of SSS to DVB in the starting mixture to 0.03, 0.07, 0.16, 0.27, and 0.42, the PDVB-SSS materials with water-droplet contact angles at 86, 68, 62, 40, and 29° were obtained, respectively (fig. S38). As shown in table S9, more SSS in the PDVB-SSS materials have lower CO conversion, showing the CO conversion at 34.9, 28.8, and 25.2% over the PDVB-SSS with SSS/DVB at 0.03, 0.07, and 0.16, respectively. For more hydrophilic samples of PDVB-SSS with SSS/DVB at 0.27 and 0.42, the CO conversions at 5.9-6.2% were obtained, supporting the importance of hydrophobic promoter.

These data confirm that the hydrophobic PDVB is crucial for realizing the desired performances, and addition of hydrophilic individual to the promoter obviously reduced the activity and/or selectivity in CoMn/C catalyzed syngas conversion. However, it is also notable that the promoter with different compositions might also lead to other unknown factors to influence the catalysis, therefore the effect of wettability on catalytic performances can only be compared within the promoters with the same components (e.g. PDVB/SiO₂ or PDVB-SSS).

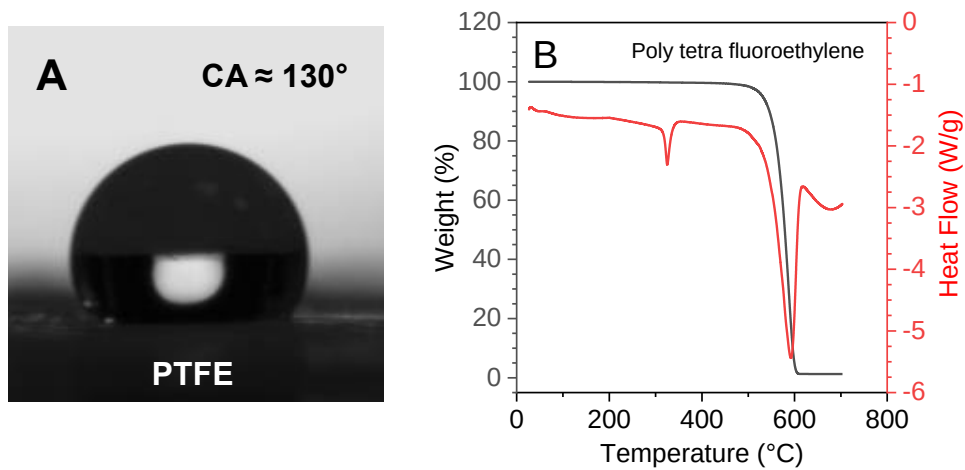


Figure S39. (A) Water-droplet contact angle and (B) TG/DSC profiles of PTFE.

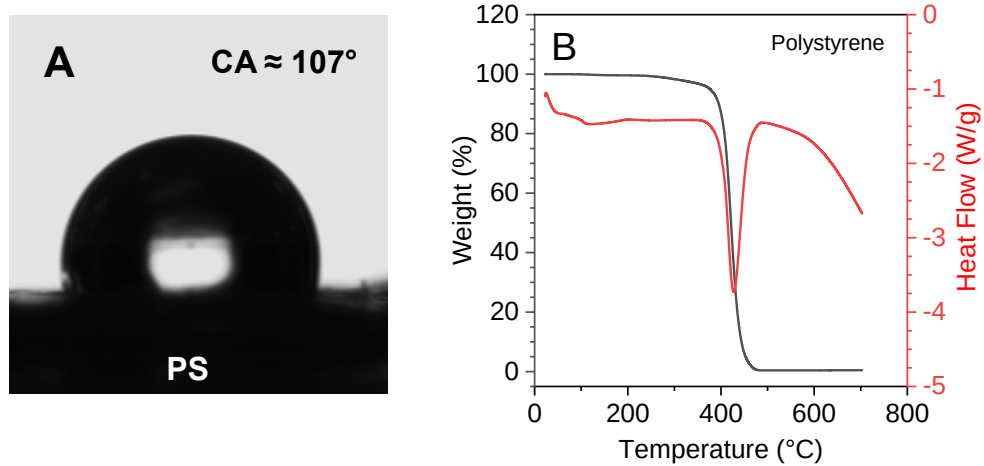


Figure S40. (A) Water-droplet contact angle and (B) TG/DSC profiles of PS.

Table S1. Catalytic data of various catalysts in syngas conversion.^[a]

Catalyst	CO	CO ₂	Selectivity (C%, CO ₂ -free)				o/p			
	Conv. (%)	Sel. (%)	CH ₄	C ₂₋₄ ⁰	C ₂₋₄ ⁼	C ₅₊ & C- oxy.	C ₂	C ₃	C ₄	C ₂ -C ₄
CoMnC	32.2	48.1	3.3	2.4	60.8	33.5	15	36	30	25
CoMnC/PDVB	63.5	46.3	4.8	6.9	71.4	16.9	3	20	25	10
CoMnC ^[b]	15.4	44.7	9.7	6.0	72.3	12.0	6	24	21	12
CoMnC/PDVB ^[b]	50.7	47.4	6.1	4.2	73.8	15.9	9	31	24	18
CoMnC ^[c]	5.5	42.8	15.7	5.5	70.7	8.1	8	29	19	13
CoMnC/PDVB ^[c]	19.4	46.2	10.9	4.2	72.2	12.7	12	26	19	17
CoMnC/PDVB-EA	88.7	49.3	4.2	8.7	53.4	33.7	1	12	11	6
CoMnC/PDVB-DMF	92.4	48.5	5.3	13.4	37.6	43.7	0.4	4	4	3
CoMnC/SiO ₂ -Me	74.7	49.2	7.6	15.5	45.5	31.4	1	7	6	3
CoMnC/Gra	53.0	47.5	6.8	4.5	67.3	21.4	7	26	19	15
Used-CoMnC ^[d]	26.2	48.4	7.7	5.9	54.9	31.5	4	16	16	9
CoMn@Si	23.7	4.3	27.9	5.8	25.9	40.4	1	9	8	4
CoMn@Si-c	22.2	4.5	28.9	6.2	26.5	38.4	1	9	7	4

[a] Reaction conditions: 1.0 g of CoMnC or 1.0 g of CoMnC physically mixed with 1.0 g of promoter, or 1.0 g of silica modified CoMn catalysts, H₂/CO at 2, 1800 mL·h⁻¹·g⁻¹, 0.1 MPa, 250 °C, powder-mixing for CoMnC and promoters.

[b] 3600 mL·h⁻¹·g⁻¹, 270 °C.

[c] 7200 mL·h⁻¹·g⁻¹, 270 °C.

[d] The used-CoMnC was obtained by separating the PDVB component of the used CoMnC/PDVB catalyst and reused in the next run.

Table S2. Surface area and pore volume of various samples by N₂ sorption measurement.

Catalyst	S _{BET} (m ² g ⁻¹)	V (cm ³ g ⁻¹)
SiO ₂ -Me	278.2	0.67
PDVB-EA	488.2	0.74
PDVB-DMF	623.3	1.98
PDVB	<5	-

Table S3. Distributions of hydrocarbon products in syngas conversion over the CoMnC and CoMnC/PDVB catalysts.^[a]

Catalyst	Selectivity (C%, CO ₂ -free)													
	CH ₄	C ₂ ⁰	C ₂ ⁼	C ₃ ⁰	C ₃ ⁼	C ₄ ⁰	C ₄ ⁼	C ₅ ⁰	C ₅ ⁼	C ₆ ⁰	C ₆ ⁼	C ₇ ⁰	C ₇ ⁼	C ₈ & C ₈₊
CoMnC	3.3	0.9	12.9	0.9	30.8	0.6	17.1	0.5	12.2	0.3	7.5	0.2	5.1	7.7
CoMnC/P DVB	4.8	4.1	13.0	1.9	37.4	0.9	21.0	0.3	8.5	0.3	4.4	0.1	1.7	1.6

[a] Reaction conditions: 1.0 g of CoMnC or 1.0 g of CoMnC physically mixed with 1.0 g of promoter, H₂/CO at 2, 1800 mL·h⁻¹·g⁻¹, 0.1 MPa, 250 °C, powder-mixing for CoMnC and promoter.

Table S4. Catalytic data of the reaction with different feed gas over the CoMnC/PDVB catalyst.^[a]

H ₂ /CO/CO ₂ /Ar	CO	CO ₂	Selectivity (C%, CO ₂ -free)				o/p			
	Conv. (%)	Sel. (%)	CH ₄	C ₂₋₄ ⁰	C ₂₋₄ ⁼	C ₅₊ & C- oxy.	C ₂	C ₃	C ₄	C ₂ -C ₄
2	63.5	46.3	4.8	6.9	71.4	16.9	3	20	25	10
3	80.2	41.3	12.4	4.6	54.8	28.2	5	21	17	12
4	88.7	38.7	15.8	6.0	49.6	28.6	3	17	14	8
3.6 ^[b]	61.7	23.7	16.7	7.0	59.5	16.8	3	21	17	8

[a] Reaction conditions: 1.0 g of CoMnC physically mixed with 1.0 g of PDVB, H₂/CO at 2, GHSV at 1800 mL·h⁻¹·g⁻¹, 0.1 MPa, 250 °C, powder-mixing for CoMnC and PDVB.

[b] The component feed gas is H₂/CO/CO₂/Ar at 68/19/10/3.

Note: We optimized the molar ratio of H₂/CO in the feed gas, reducing the CO₂ selectivity to 38.7% with H₂/CO feed ratio as high as 4. Adjusting the H₂/CO ratio to 3.6 and introducing a small amount of CO₂ in the syngas feed could remarkably reduce the CO₂ selectivity in the products to 23.7%. These data confirm that CO₂ in the feed is necessary for further reduction of the CO₂ selectivity in the products.

Table S5. Data showing the performances of various catalysts reported in literature.

Catalyst	T (°C)	P (MPa)	H ₂ /CO (molar)	CO Conv. (%)	CO ₂ Sel. (%)	Selectivity (C%, CO ₂ -free)				Yield of C ₂ -C ₄ olefins (%)	Ref.
						CH ₄	C ₂₋₄ ⁰	C ₂₋₄ ⁼	C ₅₊		
CoMnC/PDVB	250	0.1	2	63.7	49.0	4.7	6.8	71.2	17.3	23.1	TW ^[a]
CoMn	250	0.1	2	31.8	47.3	5.0	2.0	60.8	32.2	10.2	8
FeN-Mn/CNT	300	0.5	1	11.8	36.1	23.6	8.3	43.9	24.2	3.3	42
6wt%Mn/Fe ₃ O ₄	320	1.0	1	41.5	37.8	9.7	6.5	60.1	23.7	15.5	43
Fe-Na-S/ α -Al ₂ O ₃	340	2.0	1	80.0	50.0	11.0	6.0	53.0	30.0	21.2	14
Fe/N-Gra	340	0.5	1	21.1	35.1	21.4	9.2	49.6	19.8	6.8	44
ZnCrO _x /MSAPO-34	400	2.5	2.5	17.0	45.0	2	14.0	80.0	4.0	7.5	15
ZnO-ZrO ₂ /SAPO-34	400	1.0	2	11.0	45.0	4.0	18.0	74.0	4.0	4.5	16
ZnO-ZrO ₂ /SSZ-13	400	3.0	2	29.0	45.0	2	16.0	77.0	4.0	12.3	45
ZrO ₂ -In ₂ O ₃ /SAPO-34	400	2.0	1	13.1	40.0	6.2	12.5	79.9	1.3	6.3	46

[a] This work.

Table S6. Catalytic data of catalysts with different mixing manners of the CoMnC and PDVB.^[a]

Mixture manner	CO	CO ₂	Selectivity (C%, CO ₂ -free)				o/p			
	Conv. (%)	Sel. (%)	CH ₄	C ₂₋₄ ⁰	C ₂₋₄ ⁼	C ₅₊ & C- oxy.	C ₂	C ₃	C ₄	C ₂ -C ₄
Dual-bed	29.1	46.3	7.8	2.4	61.2	28.6	16	34	27	25
10-20 mesh	47.7	46.6	7.5	7.8	58.6	26.1	3	13	15	7
20-40 mesh	52.2	46.9	7.0	7.6	59.8	25.6	3	15	15	8
40-60 mesh	53.7	46.8	6.8	6.0	66.4	20.8	4	18	18	11
Powder-mixing	63.5	46.3	4.8	6.9	71.4	16.9	3	20	25	10

[a] Reaction conditions: 1.0 g of CoMnC and 1.0 g of PDVB, H₂/CO at 2, 1800 mL·h⁻¹·g⁻¹, 0.1 MPa, 250 °C.

Table S7. Catalytic data of the CoMnC/PDVB catalysts with different PDVB/CoMnC weight ratios of 0.5, 1, and 1.5.^[a]

PDVB/CoMnC (weight ratios)	CO	CO ₂	Selectivity (C%, CO ₂ -free)				o/p			
	Conv. (%)	Sel. (%)	CH ₄	C ₂₋₄ ⁰	C ₂₋₄ ⁼	C ₅₊ & C- oxy.	C ₂	C ₃	C ₄	C ₂ -C ₄
0.5	56.4	48.2	9.8	3.0	62.8	24.4	10	31	29	21
1	63.5	46.3	4.8	6.9	71.4	16.9	3	20	25	10
1.5	70.5	48.5	8.9	3.3	64.5	23.3	8	29	25	20

[a] Reaction conditions: 1.0 g of CoMnC physically mixed with 0.5, 1.0 and 1.5g of PDVB, H₂/CO at 2, 1800 mL·h⁻¹·g⁻¹, 0.1 MPa, 250 °C, powder-mixing for CoMnC and PDVB.

Table S8. Data showing the performances of CoMnC catalyst mixed with PDVB/SiO₂ promoters in the syngas conversion.^[a]

Weight ratios of PDVB/SiO ₂ in the promoter	CO	CO ₂	Selectivity (C%, CO ₂ -free)				o/p			
	Conv. (%)	Sel. (%)	CH ₄	C ₂₋₄ ⁰	C ₂₋₄ ⁼	C ₅₊ & C- oxy.	C ₂	C ₃	C ₄	C ₂ -C ₄
1/9	34.8	48.6	4.5	3.0	62.7	29.8	11	30	24	21
4/6	52.4	48.0	4.6	3.3	65.4	26.7	10	30	23	20
7/3	65.9	46.9	4.8	3.5	63.2	28.5	8	27	22	18

[a] Reaction conditions: 1.0 g of CoMnC physically mixed with 1.0 g of promoter (80-100 mesh), H₂/CO at 2, 1800 mL·h⁻¹·g⁻¹, 0.1 MPa, and 250 °C.

Table S9. Data showing the performances of CoMnC catalyst mixed with PDVB-SSS promoters with different molar ratios of SSS/DVB.^[a]

Molar ratio of SSS/DVB	CO Conv. (%)	CO ₂ Sel. (%)	Selectivity (C%, CO ₂ -free)				o/p			
			CH ₄	C ₂₋₄ ⁰	C ₂₋₄ ⁼	C ₅₊ & C-oxy.	C ₂	C ₃	C ₄	C ₂ -C ₄
0	63.5	46.3	4.8	6.9	71.4	16.9	3	20	25	10
0.03	34.9	45.9	5.0	2.1	57.2	35.7	17	35	27	27
0.07	28.8	44.4	5.5	2.1	59.2	33.2	20	36	27	29
0.16	25.2	45.9	6.1	2.6	56.5	34.8	12	32	26	22
0.27	5.9	46.7	8.0	2.2	67.2	22.6	21	40	34	30
0.42	6.2	40.3	7.9	2.4	64.6	25.1	18	38	32	27

[a] Reaction conditions: 1.0 g of CoMnC physically mixed with 1.0 g of PDVB modified by different amounts of sulfonate, H₂/CO at 2, 1800 mL·h⁻¹·g⁻¹, 0.1 MPa, 250 °C, and powder-mixing.

Table S10. Data characterizing the performances of the CoMnC catalyst mixed with different hydrophobic materials in FTO reaction.^[a]

Catalyst	CO	CO ₂	Selectivity (C%, CO ₂ -free)				o/p			
	Conv. (%)	Sel. (%)	CH ₄	C ₂₋₄ ⁰	C ₂₋₄ ⁼	C ₅₊ & C- oxy.	C ₂	C ₃	C ₄	C ₂ -C ₄
CoMnC/PTFE	37.4	44.4	12.5	4.1	59.7	23.7	6	27	24	14
CoMnC/PS	10.7	46.7	37.8	10.0	37.8	14.4	2	7	9	4

[a] Reaction conditions: 1.0 g of CoMnC physically mixed with 1.0 g of promoter catalysts (40-60 mesh), H₂/CO at 2, 1800 mL·h⁻¹·g⁻¹, 0.1 MPa, and 250 °C.

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